# Practical No.1

---

**1a. Aim:** Create a simple calculator application using servlet. **Date: / /**

**CODE:**

**index.html**

```
<html><head>
<title>Calculator App</title></head>
<body>
<form action="CalServlet" >
Enter First Number <input type="text" name="txtN1" ><br>
<select name="op>
<option value="+"> + </option>
<option value="-"> - </option>
<option value="*"> * </option>
<option value="/"> / </option> </select><br>
Enter Second Number <input type="text" name="txtN2" ><br>
<input type="reset">
<input type="submit" value="Calculate" >
</form></body>
</html>
```

---

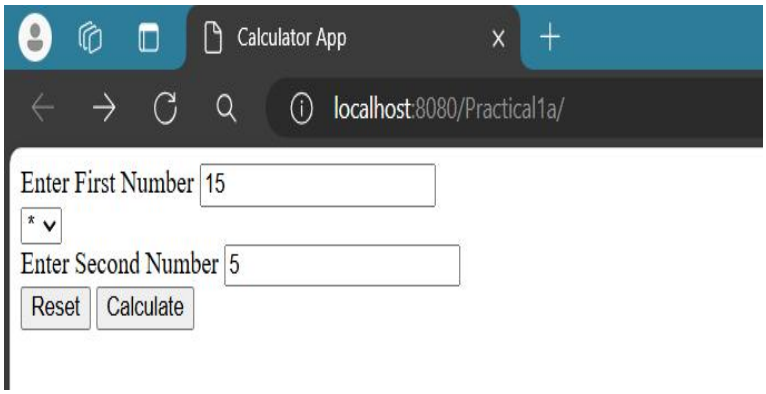
**CalServlet.java**

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

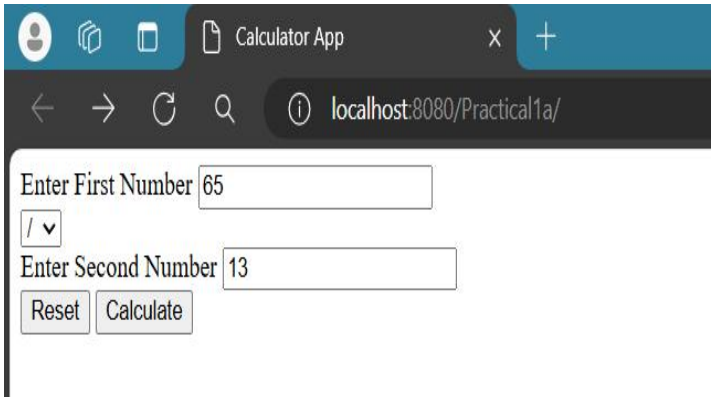
```
public class CalculatorServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
 throws ServletException, IOException
{
 response.setContentType("text/html;charset=UTF-8");
 PrintWriter pw = response.getWriter();
 int n1 = Integer.parseInt(request.getParameter("txtN1"));
 int n2 = Integer.parseInt(request.getParameter("txtN2"));
 int result =0;
 String opr=request.getParameter("op");
 if(opr.equals("+")) result=n1+n2;
 if(opr.equals("-")) result=n1-n2;
 if(opr.equals("*")) result=n1*n2;
 if(opr.equals("/")) result=n1/n2;
 pw.println("<h1> Result = "+result);
 pw.println("</body></html>");
}
}
```

---

**Output:**



Servlet CalServlet
localhost:8080/Practical1a/CalServlet?txtN1=15&op=*&txtN2=5
Result = 75
Servlet CalServlet at /Practical1a


Calculator App
localhost:8080/Practical1a/
Enter First Number
65
/
Enter Second Number
13
Reset
Calculate

Servlet CalServlet
localhost:8080/Practical1a/CalServlet?txtN1=65&op=%2F&txt
Result = 5
Servlet CalServlet at /Practical1a

**Date: / /**

**1b. Aim**: Create a servlet for a login page. If the username and password are correct then it says message “Hello <username>” else a message “login failed”.

**CODE:**

**index.html**

```
<html><head><title>Login Form</title></head>
<form action="LoginServlet" >
Enter User ID<input type="text" name="txtId"><br>
Enter Password<input type="password" name="txtPass"><br>
<input type="reset"><input type="submit" value=" Click to Login " >
</form>
</html>
```

---

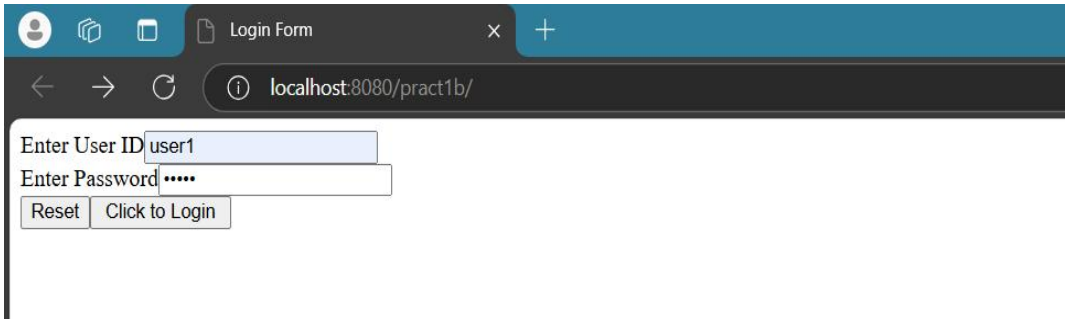
**LoginServlet.java**

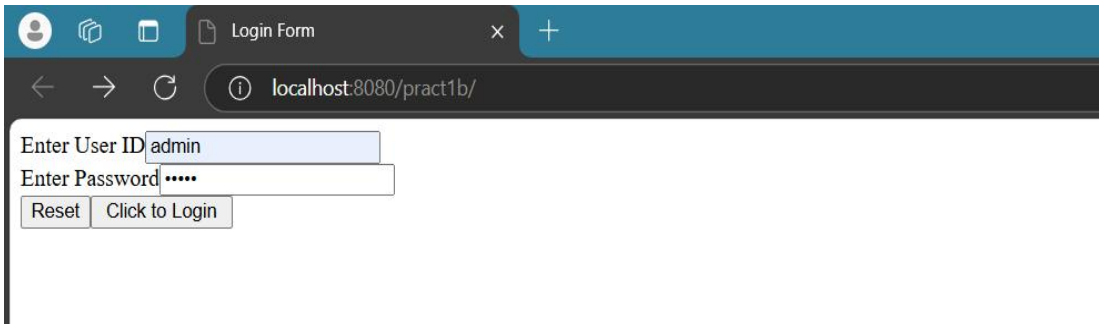
```
import java.io.*;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class LoginServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
    out.println("<html><head><title>Servlet LoginServlet</title></head>");
    String uname = request.getParameter("txtId");
    String upass = request.getParameter("txtPass");
    if(uname.equals("admin") && upass.equals("12345")){
    out.println("<body bgcolor=blue >");
    out.println("<h1> Hello!!! "+uname+"</h1>");
    } else {
```

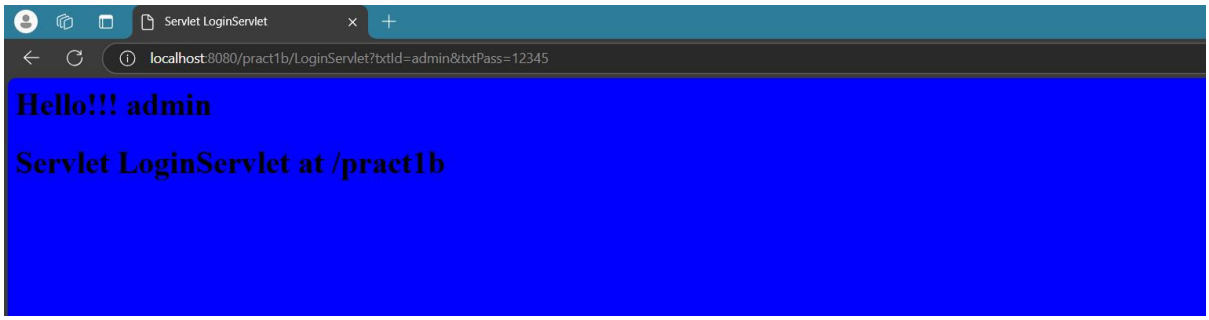
```
            out.println("<body bgcolor=red >");
            out.println("<h1> Login Fail !!! </h1>");
        }
        out.println("</body></html>");
    }
}
```

---

**Output:**





**Date: / /**

**1c. Aim**: Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

**CODE :**

**for Database**

```
>> create database logindb;

>> use logindb;

>> create table user (username varchar(10) PRIMARY KEY, password varchar(20), email
        varchar(20), country(10));
```

---

**index.html**

```
<html><head><title>Registration Page</title></head>

<body><form action="registerServlet" > <H1>Welcome to Registration page</H1>

Enter User Name <input type="text" name="txtUid"><br>

Enter Password <input type="password" name="txtPass"><br>

Enter Email <input type="text" name="txtEmail" ><br>

Enter Country <input type="text" name="txtCon" ><br>

<input type="reset" ><input type="submit" value="REGISTER" ></form></body></html>
```

---

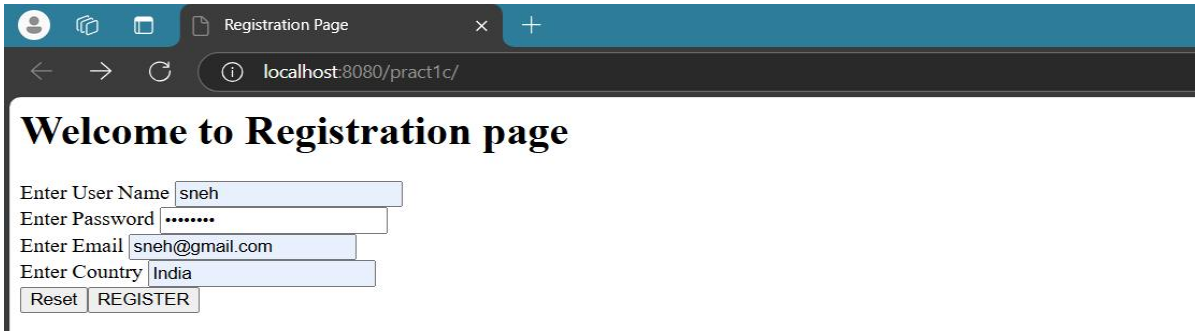
**registerServlet.java**

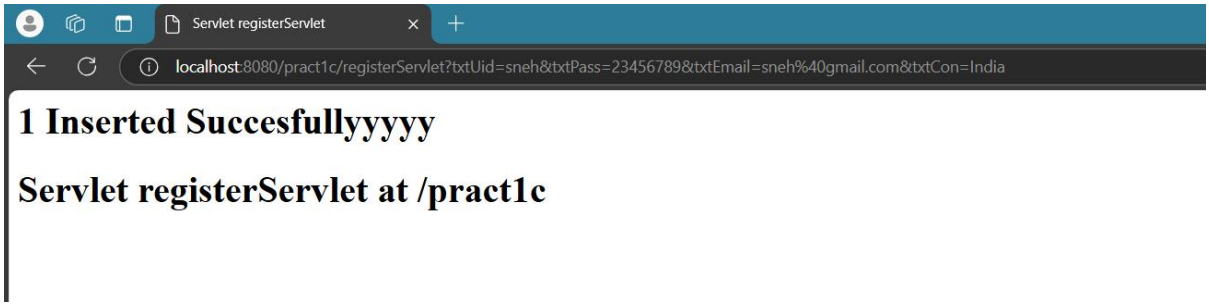
```
import java.io.*;

import javax.servlet.ServletException;

import javax.servlet.http.*;

import java.sql.*;

public class registerServlet extends HttpServlet {

protected void doGet(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

    response.setContentType("text/html;charset=UTF-8");

    PrintWriter out = response.getWriter();
```

```
        String id = request.getParameter("txtUid");
        String ps = request.getParameter("txtPass");
        String em = request.getParameter("txtEmail");
        String co = request.getParameter("txtCon");
        try{
            Class.forName("com.mysql.jdbc.Driver");
            Connection con
=DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb","root","root");
            PreparedStatement pst = con.prepareStatement("insert into user values(?,?,?,?)");
            pst.setString(1,id);
            pst.setString(2,ps);
            pst.setString(3,em);
            pst.setString(4,co);
            int row = pst.executeUpdate();
            out.println("<h1>"+row+ " Inserted Succesfullyyyyy");
        } catch (Exception e) {out.println(e); }
    }
```

---

**Output :**



# Practical No.2

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**Date : / /**

**2a. Aim :** Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered "servlet" as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.

**CODE :**

**index.html**

```
<html><head><title>Login Form</title></head>
<form action="LoginServlet" >
Enter User ID<input type="text" name="txtId"><br>
Enter Password<input type="password" name="txtPass"><br>
<input type="reset"><input type="submit" value=" Click to Login " >
</form></html>
```

---

**LoginServlet.java**

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
public class LoginServlet extends HttpServlet {
  protected void doGet(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
       PrintWriter out = response.getWriter();
       out.println("<html><head>");
```

```
        out.println("<title>Servlet LoginServlet</title></head>");
        String uname = request.getParameter("txtId");
        String upass = request.getParameter("txtPass");
        if(uname.equals("admin") && upass.equals("servlet")) {
            RequestDispatcher rd = request.getRequestDispatcher("WelcomeServlet");
            rd.forward(request, response);
        } else {
            out.println("<h1> Login Fail !!! </h1>");
            RequestDispatcher rd = request.getRequestDispatcher("index.html");
            rd.include(request, response);
        }
        out.println("</body>");
        out.println("</html>");
    }
}
```

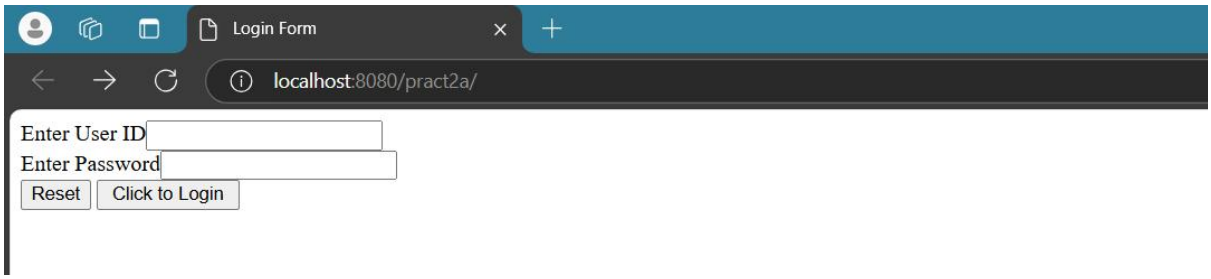
---

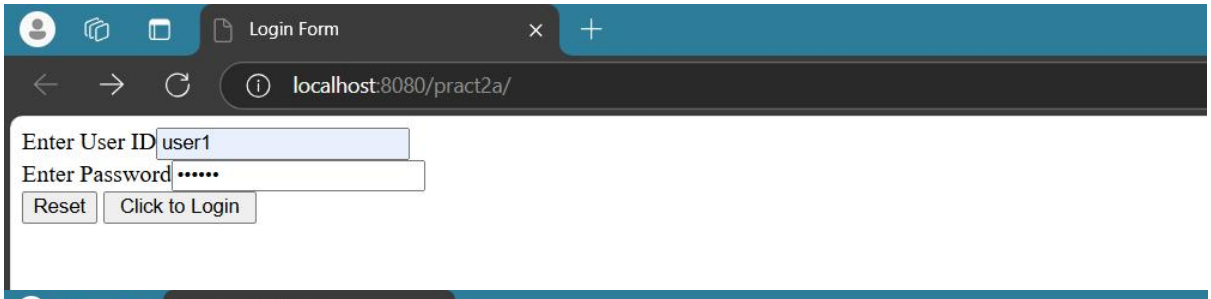
**WelcomeServlet.java**

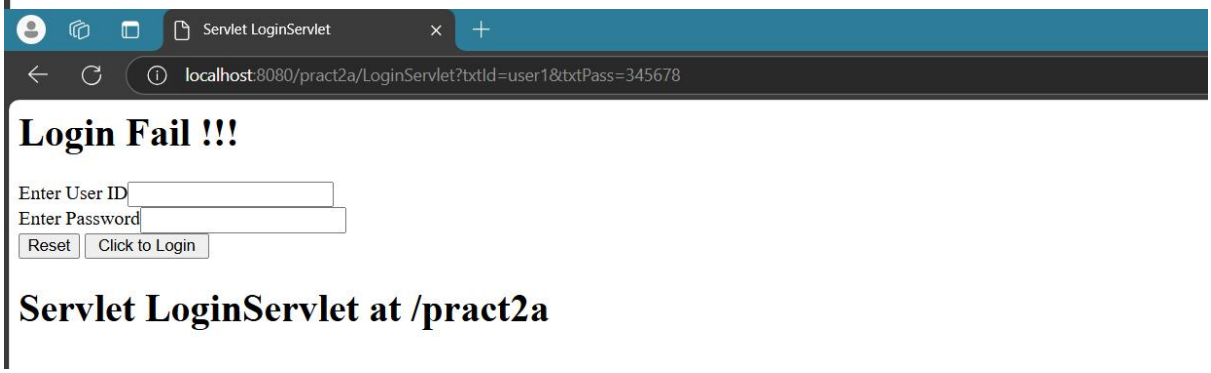
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
public class WelcomeServlet extends HttpServlet {
  protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        out.println("<html><head>");
```

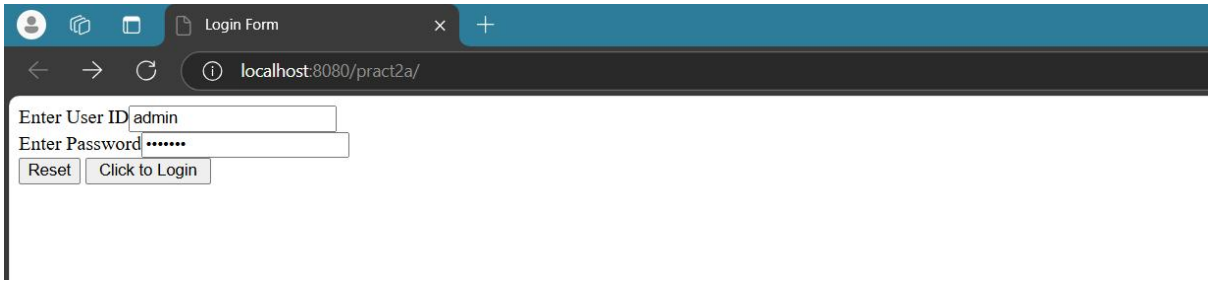
```
        out.println("<h1>Login Successful!!</h1>");
    }
}
```

**Output :**



**Date : / /**

**2b. Aim :** Create a servlet that uses Cookies to store the number of times a user has visited servlet.

**CODE :**

**index.html**

```
<html>
<head><title>Cookie Demo</title></head>
<body><form action="count" >
Enter Your Name <input type="text" name="t1"><br>
<input type="submit" value="Click to Enter ">
</form></body></html>
```

---

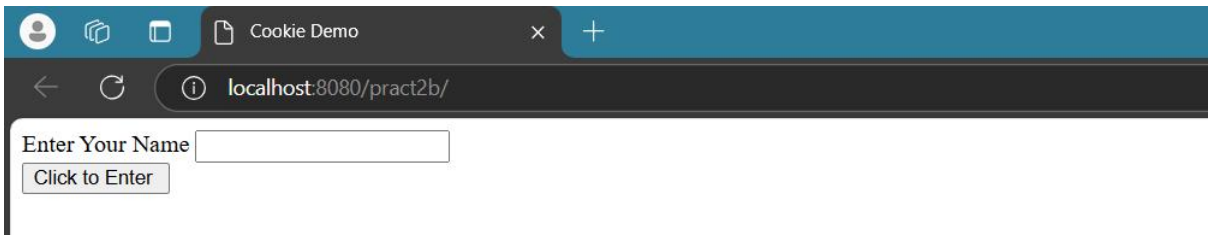
**count.java**

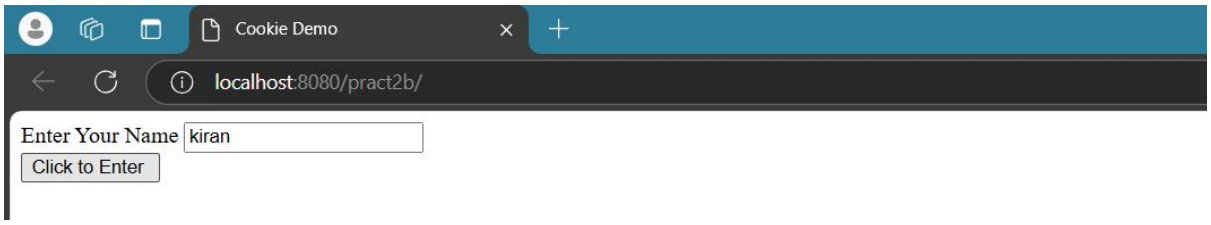
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Cookie;
public class count extends HttpServlet {
static int count=0,c2=0;
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
      String name=request.getParameter("t1");
      Cookie c1=new Cookie("count",String.valueOf(count));
      c2=Integer.parseInt(c1.getValue());
      if (c2==0) {
```

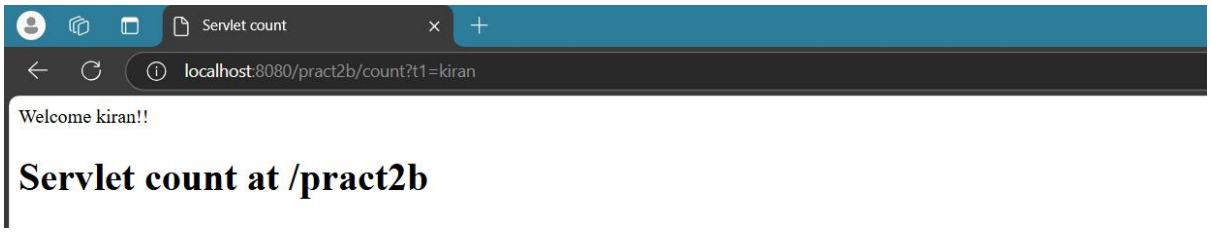
```
            out.println("Welcome = "+name);
            count++;
        } else {
            c1=new Cookie("count",String.valueOf(count));
            count++;
            out.println("Welcome = "+ name + "You have visited the page " +count + " times.");
        }
    }
}
```

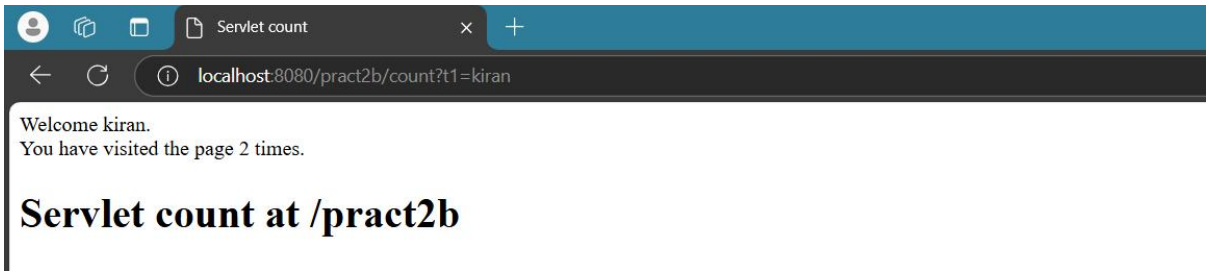
---

**Output :**



**Date :  /  /**

**2c. Aim :** Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.

**CODE :**

**index.html**

```
<html><head><title>Session Demo</title></head>
<form action="Page1" method="get" >
Enter User ID <input type="text" name="txtName"><br>
<input type="reset" ><input type="submit" >
</form></html>
```

---

**Page1.java**

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class Page1 extends HttpServlet {
   public void doGet(HttpServletRequest request, HttpServletResponse response)
     throws ServletException, IOException {
          response.setContentType("text/html;charset=UTF-8");
          PrintWriter out = response.getWriter();
          out.println("<html><head><title>Servlet Page1</title></head>");
          HttpSession hs = request.getSession(true);
          if (hs.isNew()) {
                out.println("<body bgcolor=yellow>");
                String name = request.getParameter("txtName");
                hs.setAttribute("uname", name);
                hs.setAttribute("visit", "1");
                out.println("<h1>Welcome First Time</h1>");
          } else {
```

```
            out.println("<h1>Welcome Again</h1>");
            int visit = Integer.parseInt((String)hs.getAttribute("visit"))+1;
            out.println("<h1>You Visited "+visit+"Times</h1>");
            hs.setAttribute("visit", ""+visit);
        }
    out.println("<h1>Your Session ID is "+hs.getId()+"</h1>");
    out.println("<h1>You Logged in at "+new java.util.Date(hs.getCreationTime())+"</h1>");
    out.println("<h1><a href=Page2>Click for Page 2 </a></h1>");
    out.println("<h1><a href=LogoutServlet>Click to Terminate Session </a></h1>");
    out.println("</body></html>");
  }
}
```

---

**Page2.java**

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class Page2 extends HttpServlet {
  public void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
    out.println("<html><head><title>Servlet Page2</title></head>");
    HttpSession hs = request.getSession(false);
    out.println("<h1>Welcome Again on Page No. 2</h1>");
    int visit = Integer.parseInt((String)hs.getAttribute("visit"))+1;
    out.println("<h1>You Visited "+visit+"Times</h1>");
    hs.setAttribute("visit", ""+visit);
    out.println("<h1>Your Session ID is "+hs.getId()+"</h1>");
```

```
    out.println("<h1>You Logged in at "+new java.util.Date(hs.getCreationTime())+"</h1>");
    out.println("<h1><a href=Page1>Click for Page 1 </a></h1>");
    out.println("<h1><a href=LogoutServlet>Click for Terminate Session </a></h1>");
    out.println("</body></html>");
  }
}
```

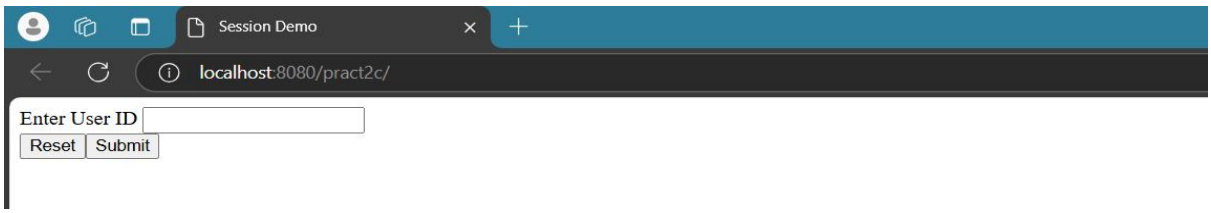
---

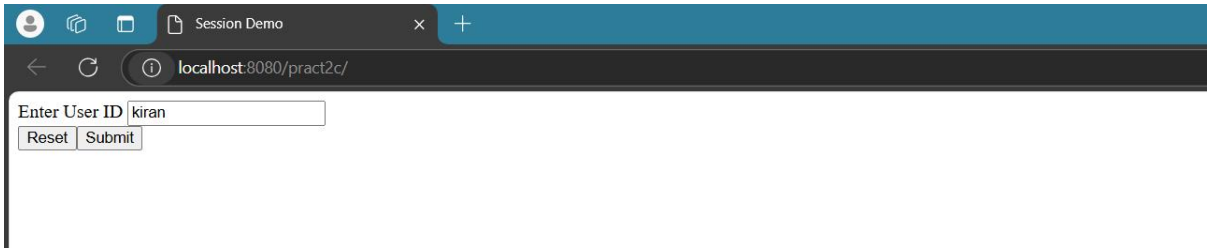
**LogoutServlet.java**

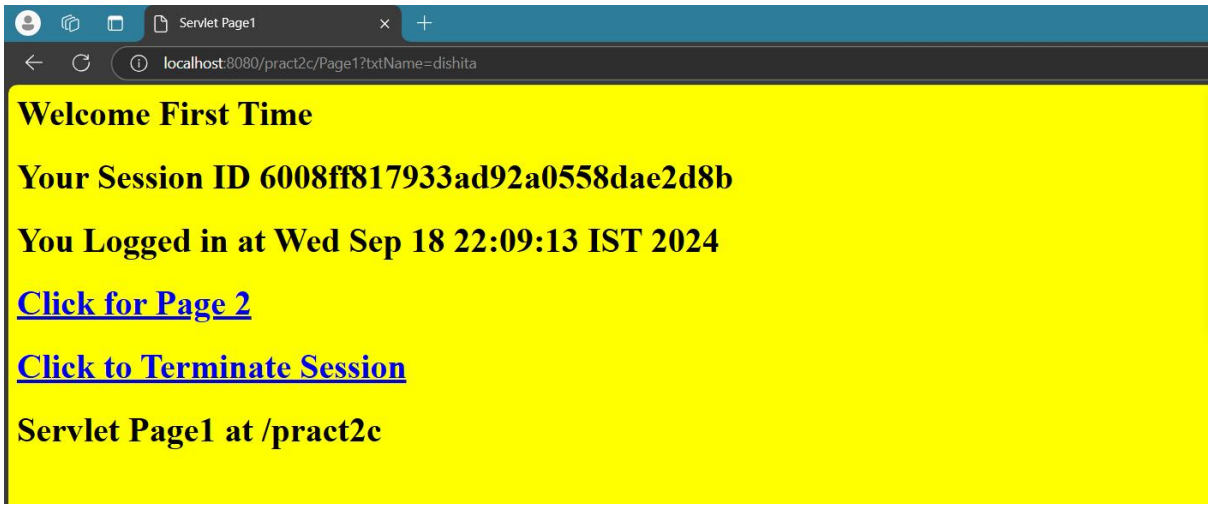
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class LogoutServlet extends HttpServlet {
  public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
    out.println("<html><head><title>Servlet LogoutServlet</title></head>");
    out.println("<body>");
    HttpSession hs = request.getSession();
    if(hs != null) hs.invalidate();
    out.println("<h1>You are Logged out now.....</h1>");
    out.println("</body></html>");
  }
}
```

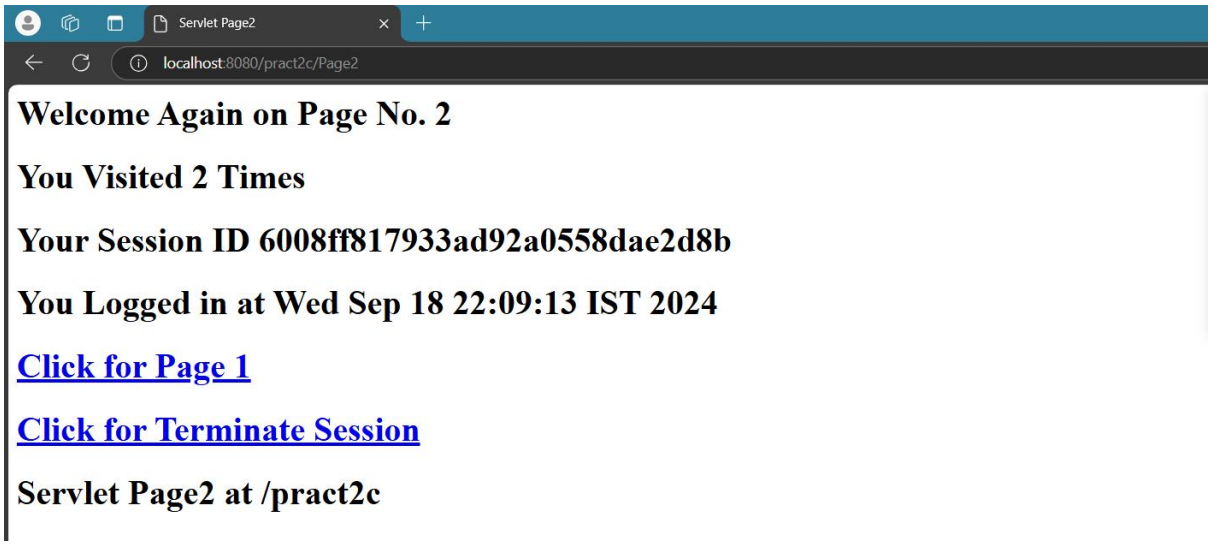
---

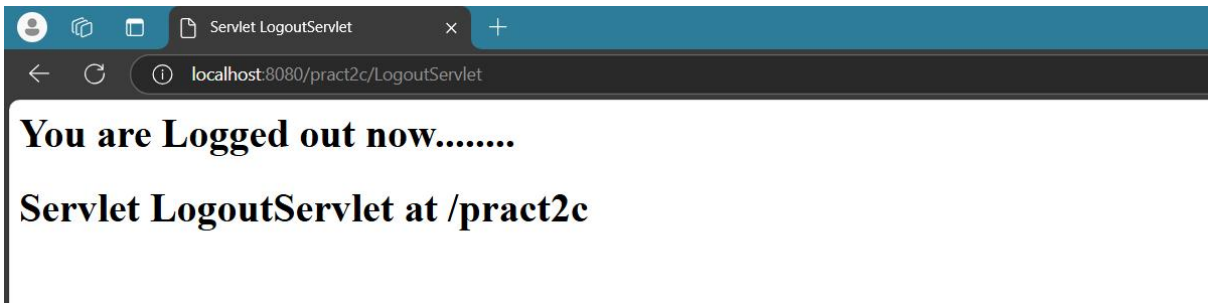
**Output :**




Session Demo
localhost:8080/pract2c/
Enter User ID kiran
Reset Submit


Servlet Page1
localhost:8080/pract2c/Page1?txtName=dishita
Welcome First Time
Your Session ID 6008ff817933ad92a0558dae2d8b
You Logged in at Wed Sep 18 22:09:13 IST 2024
Click for Page 2
Click to Terminate Session
Servlet Page1 at /pract2c


Servlet Page2
localhost:8080/pract2c/Page2
Welcome Again on Page No. 2
You Visited 2 Times
Your Session ID 6008ff817933ad92a0558dae2d8b
You Logged in at Wed Sep 18 22:09:13 IST 2024
Click for Page 1
Click for Terminate Session
Servlet Page2 at /pract2c


Servlet LogoutServlet
localhost:8080/pract2c/LogoutServlet
You are Logged out now.....
Servlet LogoutServlet at /pract2c

# Practical No.3

---

**Date : / /**

**3a. Aim :** Create a Servlet application to upload and download a file.

**CODE :**

UPLOAD FILE

**index.html**

```
<html><body>
]<form action="FileUploadServlet" method="post" enctype="multipart/form-data">
Select File to Upload:<input type="file" name="file" id="file">
Destination <input type="text" value="" name="destination"><br>
<input type="submit" value="Upload file" name="upload" id="upload">
</form></body></html>
```

---

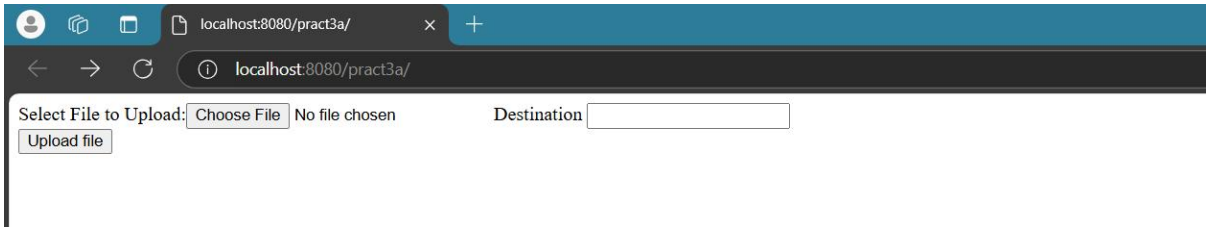
**FileUploadServlet.java**

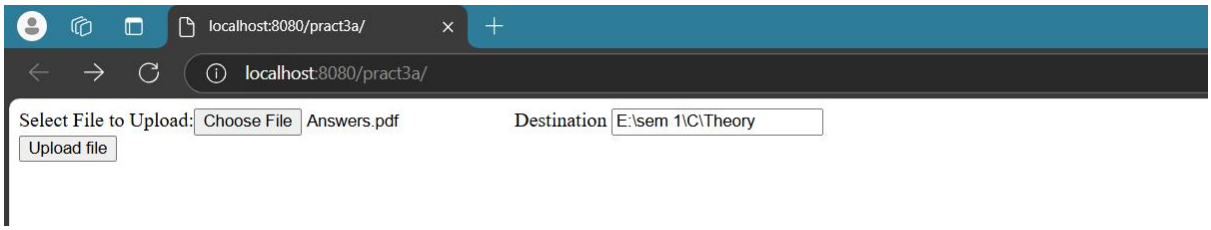
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.http.*;
@MultipartConfig
public class FileUploadServlet extends HttpServlet {
    public void doPost(HttpServletRequest request,HttpServletResponse response) throws
ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String path=request.getParameter("destination");
        Part filePart=request.getPart("file");
        String filename=filePart.getSubmittedFileName().toString();
        out.print("<br><br><hr> file name: "+filename);
```

```
        OutputStream os=null;
        InputStream is=null;
        try {
                os=new FileOutputStream(new File(path+File.separator+filename));
                is=filePart.getInputStream();
                int read=0;
                while ((read = is.read()) != -1) {
                        os.write(read);
                }
                out.println("<br>file uploaded sucessfully...!!!");
        }
        catch(FileNotFoundException e){out.print(e);}
    }
}
```

---

**Output :**



DOWNLOAD FILE :

**index.html**

```
<html><head><title>File Download Page</title>
 </head> <body> <h1>File Download Application</h1>
 Click <a href="DownloadServlet?filename=EJ Lab Manual.pdf">EJ Manual</a>
 <br/><br/>Click <a href="DownloadServlet?filename=AI.pdf">AI</a>
 </body> </html>
```

---

**(Once you create the project for downloading a file, **copy any 2 files** (pdf, docx) that already exists on your pc and paste both the files in the WEB PAGES folder of your project. **Right Click on WEB PAGES -> paste** and then write your own file name in the above html code in place of - "EJ Lab Manual.pdf" and "AI.pdf")

Example : If 1 of your file names is 'EJ.pdf' then, on the 3rd line write "filename=EJ.pdf"

---

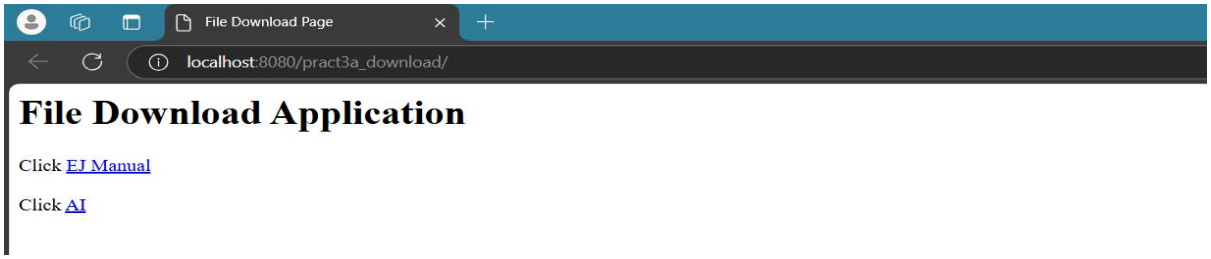
**DownloadServlet.java**

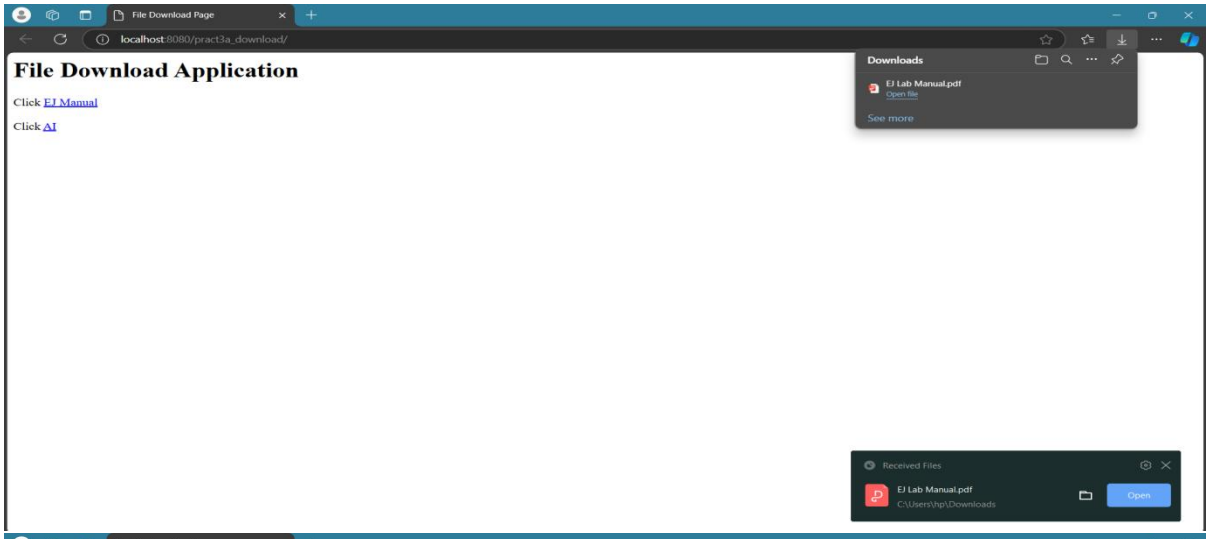
```
import java.io.InputStream;
import javax.servlet.ServletContext;
import javax.servlet.ServletOutputStream;
public class DownloadServlet extends HttpServlet {
   @Override
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
      response.setContentType("APPLICATION/OCTET-STREAM");
      String filename = request.getParameter("filename");
      ServletContext context = getServletContext();
      InputStream is = context.getResourceAsStream("/" + filename);
      PrintWriter out = response.getWriter();
      response.setHeader("Content-Disposition", "attachment; filename=\"" + filename + "\"");
      int i;
      while ((i = is.read()) != -1) {
            out.write(i);
```

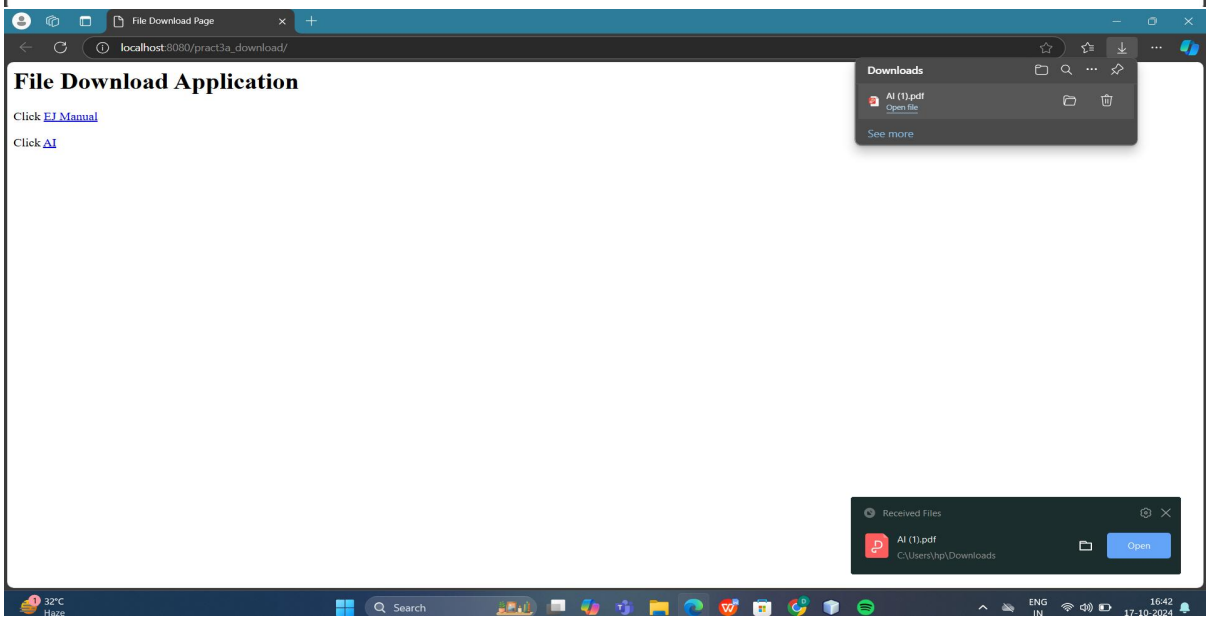
```
        }
        is.close();
        out.close();
    }
}
```

---

**Output :**



**Date : / /**

**3b. Aim :** Develop Simple Servlet Question Answer Application using Database.

**CODE :**

**for Database**

```
>> create database qadb;

>> use qadb;

>> create table quiz (qno varchar(5) PRIMARY KEY, question varchar(100), op1
        varchar(50), op2 varchar(50), op3 varchar(50), op4 varchar(50), ans varchar(50));

>> insert into quiz values ('001', 'Unit of Energy is ___', 'Dozen','Kilometre', 'Joule',
        'Hertz','Joule');

>> insert into quiz values ('002', 'Following is the extension for saving a python file',
        '.jpg','.py', '.docx', '.cs','.py');

>> insert into quiz values ('003', '___ is the largest continent', 'Asia','Antarctica', 'Europe',
            'Australia','Asia');

>> insert into quiz values ('004', '___ is the smallest memory unit', 'bit','byte', 'Kilo Byte',
            'Gigabyte','bit');

>> insert into quiz values ('005', 'Which is the only liquid metal', 'Steel','Aluminium',
        'Magnesium', 'Mercury','Mercury');
```

---

**index.html**

```
<html><head><title>Quiz Application</title></head>

<body><h1>Welcome to Quiz Servlet </h1>

<h1><a href="QuizServlet" >CLICK TO START QUIZ</a></h1> </body></html>
```

---

**QuizServlet.java**

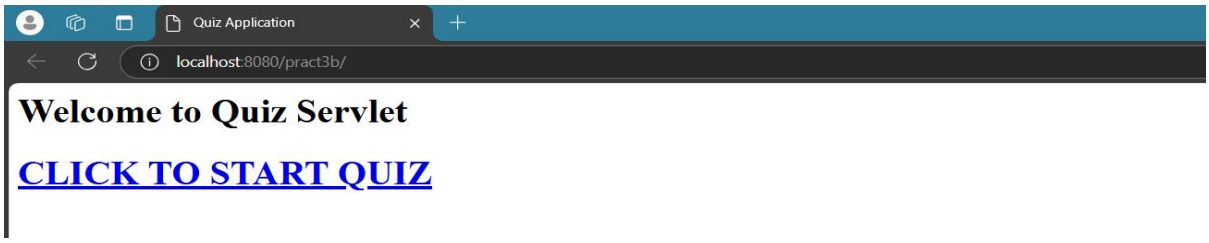
```
import java.io.*;

import java.sql.*;

import javax.servlet.*;

import javax.servlet.http.*;

public class QuizServlet extends HttpServlet {

    public void doGet(HttpServletRequest request, HttpServletResponse response)
```

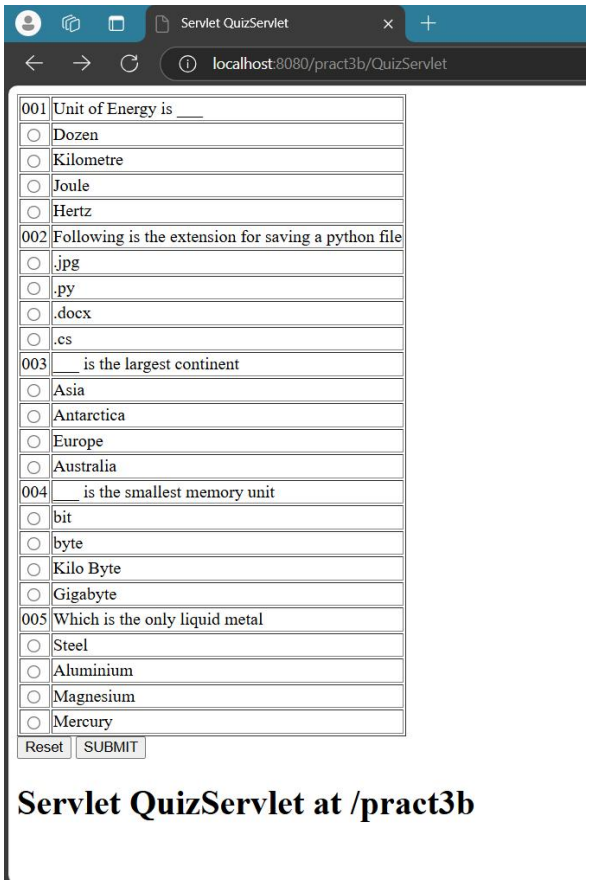
```
    throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        out.println("<form action=ShowResult >");
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con =
        DriverManager.getConnection("jdbc:mysql://localhost:3306/qadb","root","root");
            Statement stmt = con.createStatement();
            ResultSet res = stmt.executeQuery("select * from quiz");
            out.println("<table border=1 >");
            int qno=0;
            while(res.next()) {
                qno++;
                out.println("<tr><td>"+res.getString(1)+"</td>");
                out.println("<td>"+res.getString(2)+"</td></tr>");
                out.println("<tr><td><input type=radio name="+qno+"
            value="+res.getString(3)+"></td><td>"+res.getString(3)+"</td></tr>");
                out.println("<tr><td><input type=radio name="+qno+"
            value="+res.getString(4)+"></td><td>"+res.getString(4)+"</td></tr>");
                out.println("<tr><td><input type=radio name="+qno+"
            value="+res.getString(5)+"></td><td>"+res.getString(5)+"</td></tr>");
                out.println("<tr><td><input type=radio name="+qno+"
            value="+res.getString(6)+"></td><td>"+res.getString(6)+"</td></tr>");
            }
        } catch (Exception e) {out.println(e);}
        out.println("</table>");
        out.println("<input type=reset >");
        out.println("<input type=submit value=SUBMIT >");
        out.println("</form>");
    }
}
```

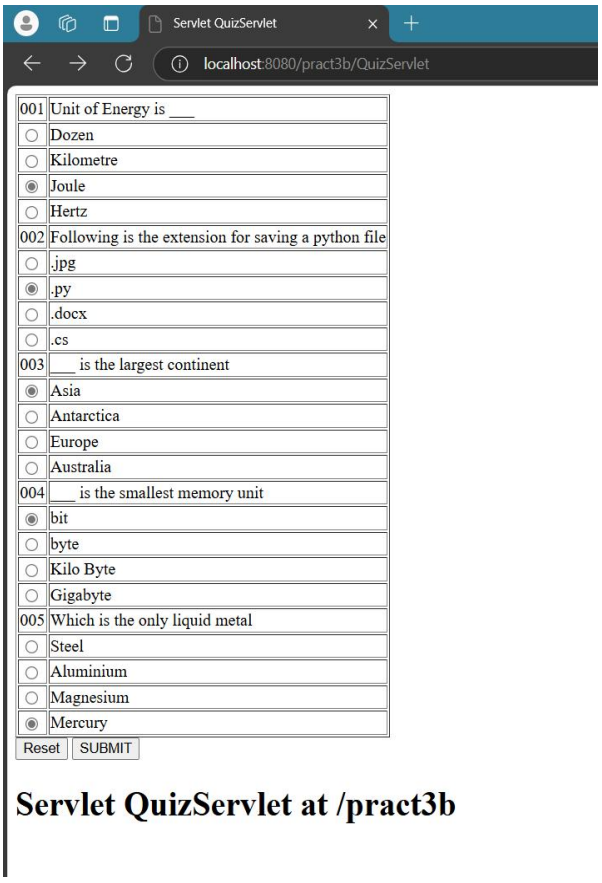
**ShowResult.java**

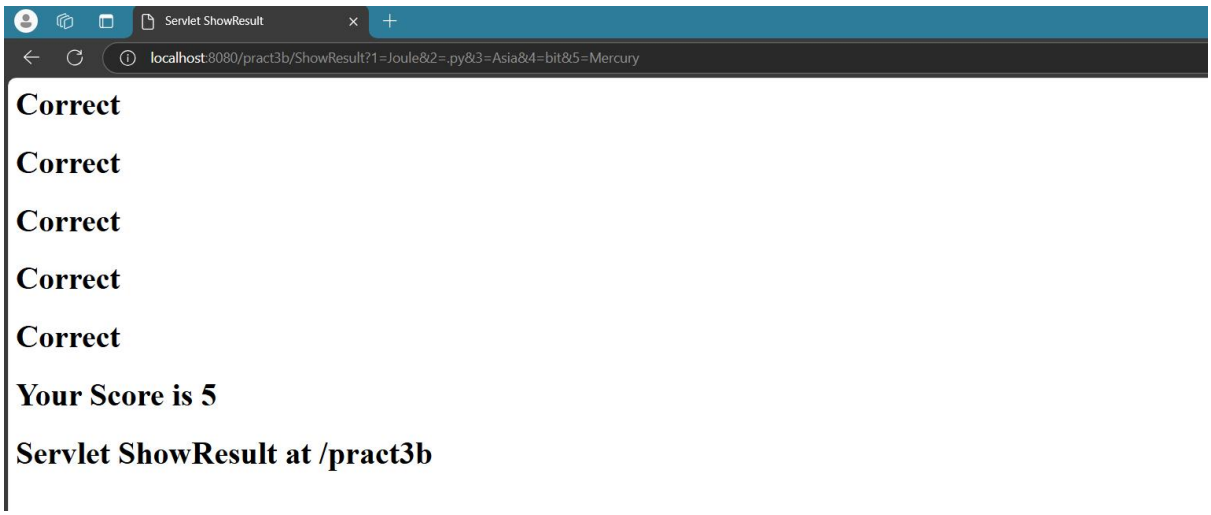
```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class ShowResult extends HttpServlet {
   public void doGet(HttpServletRequest request, HttpServletResponse response)
     throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try {
                Class.forName("com.mysql.jdbc.Driver");
                Connection con =
        DriverManager.getConnection("jdbc:mysql://localhost:3306/qadb","root","root");
                Statement stmt = con.createStatement();
                ResultSet res = stmt.executeQuery("select ans from quiz");
                int count =0, qno=0;
                while (res.next()) {
                        if (res.getString(1).equals(request.getParameter(""+(++qno)))) {
                                count++;
                                out.println("<h1>Correct </h1>");
                        }
                        else {
                                out.println("<h1>Incorrect </h1>");
                        }
                }
                out.println("<h1>Your Score is "+count+" </h1>");
        }catch (Exception e) {out.println(e);}
   }
}
```

**Output :**


Quiz Application
localhost:8080/pract3b/
Welcome to Quiz Servlet
CLICK TO START QUIZ


Servlet QuizServlet
localhost:8080/pract3b/QuizServlet
001 Unit of Energy is ___
Dozen
Kilometre
Joule
Hertz
002 Following is the extension for saving a python file
.jpg
.py
.docx
.cs
003 ___ is the largest continent
Asia
Antarctica
Europe
Australia
004 ___ is the smallest memory unit
bit
byte
Kilo Byte
Gigabyte
005 Which is the only liquid metal
Steel
Aluminium
Magnesium
Mercury
Reset
SUBMIT
Servlet QuizServlet at /pract3b


Servlet QuizServlet
localhost:8080/pract3b/QuizServlet
001 Unit of Energy is ___
Dozen
Kilometre
Joule
Hertz
002 Following is the extension for saving a python file
.jpg
.py
.docx
.cs
003 ___ is the largest continent
Asia
Antarctica
Europe
Australia
004 ___ is the smallest memory unit
bit
byte
Kilo Byte
Gigabyte
005 Which is the only liquid metal
Steel
Aluminium
Magnesium
Mercury
Reset
SUBMIT
Servlet QuizServlet at /pract3b


Servlet ShowResult
localhost:8080/pract3b/ShowResult?1=Joule&2=.py&3=Asia&4=bit&5=Mercury
Correct
Correct
Correct
Correct
Correct
Your Score is 5
Servlet ShowResult at /pract3b

# Practical No.4

---

**Date : / /**

**4a. Aim :** Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.

**CODE :**

**<u>index.jsp</u>**

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html><head><title>JSP Page</title></head><body>
<h1>Use of Intrinsic Objects in JSP</h1><h1>Request Object </h1>
Query String <%=request.getQueryString() %><br>
Context Path <%=request.getContextPath() %><br>
Remote Host <%=request.getRemoteHost() %><br>
<h1>Response Object </h1>
Character Encoding Type <%=response.getCharacterEncoding() %><br>
Content Type <%=response.getContentType() %><br>
Locale <%=response.getLocale() %><br>
<h1>Session Object </h1> ID <%=session.getId() %><br>
Creation Time <%=new java.util.Date(session.getCreationTime()) %><br>
Last Access Time<%=new java.util.Date(session.getLastAccessedTime()) %><br>
</body></html>
```

---

**Output :**

**Date : / /**

**4b. Aim :** Develop a simple JSP application to pass values from one page to another with validations.

(Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).

**CODE :**

**index.html**

```
<!DOCTYPE html>
<html><head><title>Form Page</title><script type="text/javascript">
    function validateForm() {
      var name = document.forms["myForm"]["name"].value;
      var age = document.forms["myForm"]["age"].value;
      var email = document.forms["myForm"]["email"].value;
      var hobbies = document.forms["myForm"]["hobbies"];
      var gender = document.forms["myForm"]["gender"];
      if (name == "") {
        alert("Name must be filled out");
        return false; }
      if (age == "" || isNaN(age) || age < 1 || age > 100) {
        alert("Please enter a valid age");
        return false;  }
      if (!validateHobbies(hobbies)) {
        alert("Please select at least one hobby");
        return false; }
      if (!validateGender(gender)) {
        alert("Please select your gender");
        return false;  }
      if (email == "" || !email.match(/^\S+@\S+\.\S+$/)) {
        alert("Please enter a valid email");
        return false;
      } return true; }
```

```
        function validateHobbies(hobbies) {
            for (var i = 0; i < hobbies.length; i++) {
                if (hobbies[i].checked) {
                    return true;
                }
            } return false; }
        function validateGender(gender) {
            for (var i = 0; i < gender.length; i++) {
                if (gender[i].checked) {
                    return true; }
            } return false;
        } </script>
</head><body> <h2>Enter Your Details</h2>
   <form name="myForm" action="Validate.jsp" onsubmit="return validateForm()"
method="post">
        Name: <input type="text" name="name"><br><br>
        Age: <input type="text" name="age"><br><br>
        Email: <input type="text" name="email"><br><br>
        Gender: <br>
        <input type="radio" name="gender" value="Male"> Male<br>
        <input type="radio" name="gender" value="Female"> Female<br><br>
        Hobbies: <br>
        <input type="checkbox" name="hobbies" value="Reading"> Reading<br>
        <input type="checkbox" name="hobbies" value="Music"> Music<br>
        <input type="checkbox" name="hobbies" value="Sports"> Sports<br><br>
        <input type="submit" value="Submit">
    </form> </body></html>
```

---

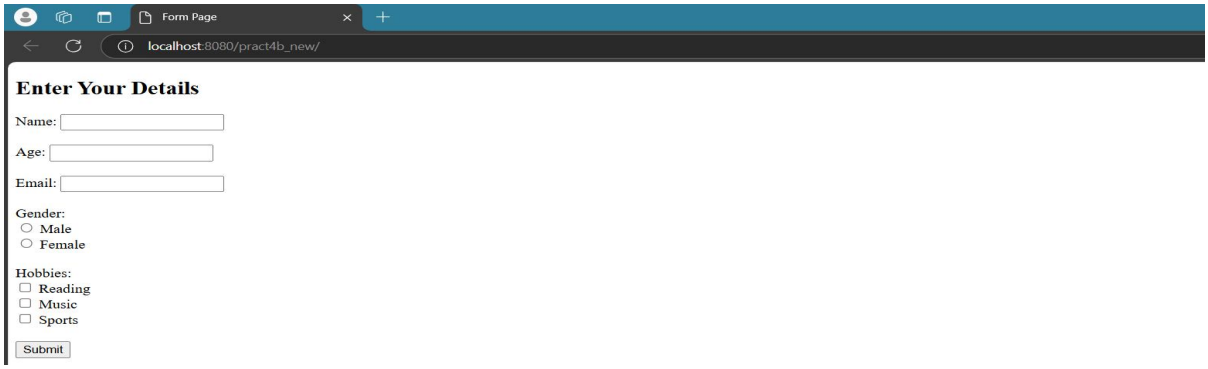
**Validate.jsp**

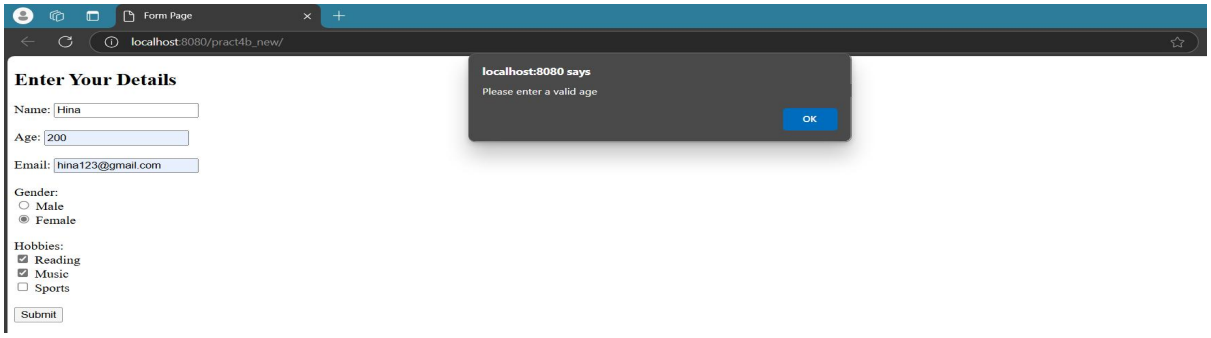
```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
```

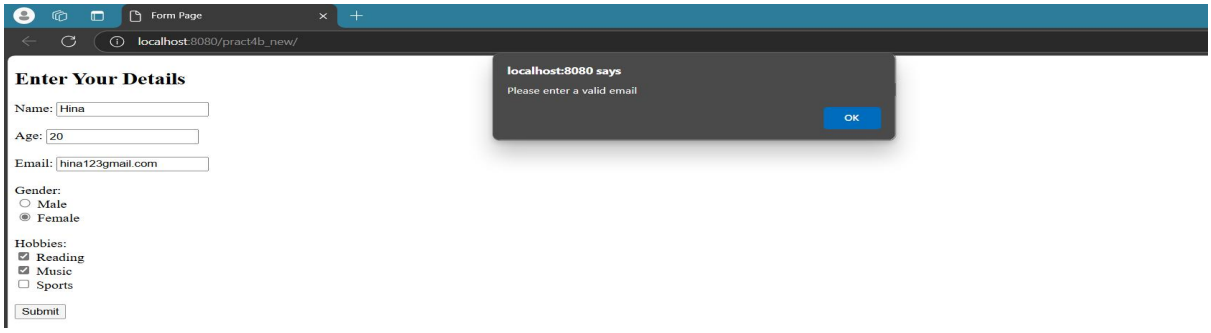
```
<html><head> <title>Result Page</title></head><body>
   <h1>Validated Successfully!!</h1> <h2>Your Submitted Information</h2>
   <% String name = request.getParameter("name");
      String age = request.getParameter("age");
      String email = request.getParameter("email");
      String gender = request.getParameter("gender");
      String[] hobbies = request.getParameterValues("hobbies");  %>
   <p><strong>Name:</strong> <%= name %></p>
   <p><strong>Age:</strong> <%= age %></p>
   <p><strong>Email:</strong> <%= email %></p>
   <p><strong>Gender:</strong> <%= gender %></p>
   <p><strong>Hobbies:</strong></p>
   <ul> <%
         if (hobbies != null) {
            for (String hobby : hobbies) {
               out.println("<li>" + hobby + "</li>");
            }
         } else {
            out.println("<li>No hobbies selected</li>");
         } %>
   </ul>
</body> </html>
```

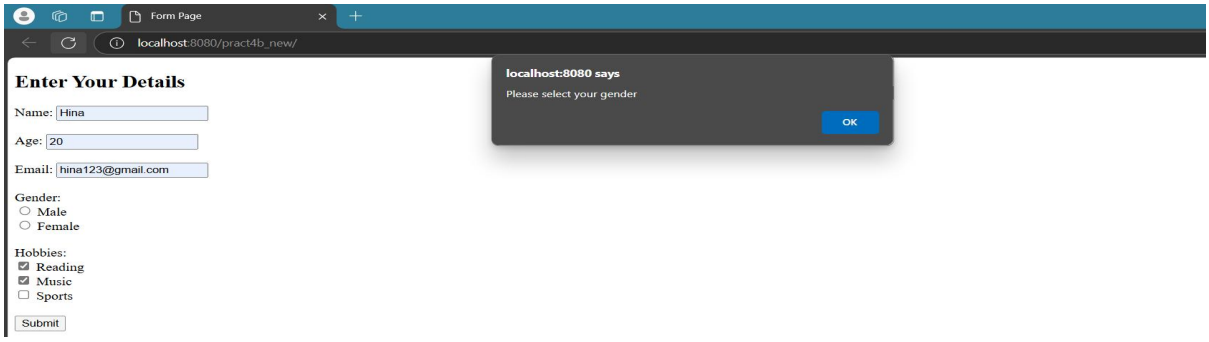
---

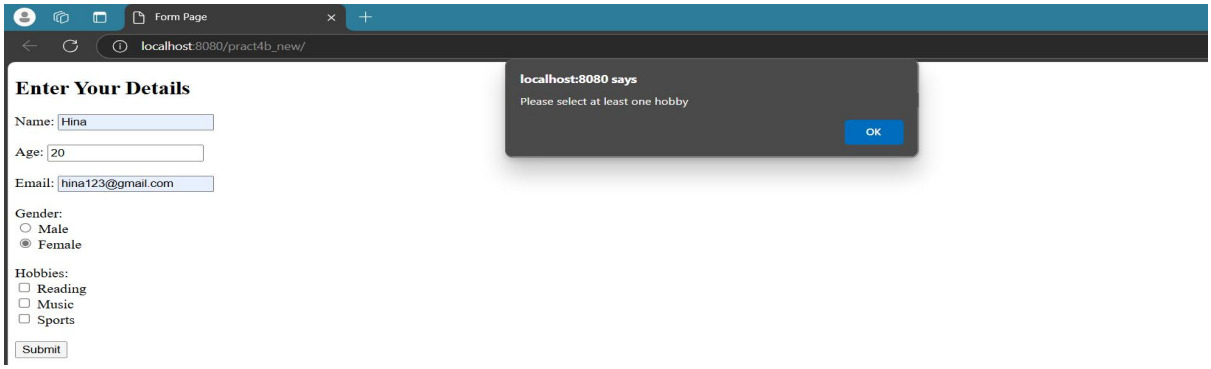
**Output :**

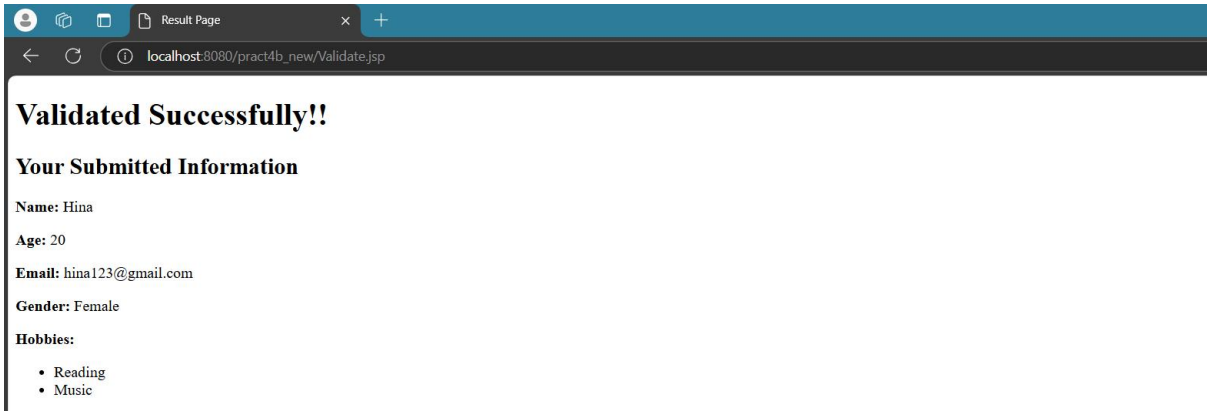



Form Page
localhost:8080/pract4b_new/
Enter Your Details
Name: Hina
Age: 200
Email: hina123@gmail.com
Gender:
Male
Female
Hobbies:
Reading
Music
Sports
Submit
localhost:8080 says
Please enter a valid age
OK


Form Page
localhost:8080/pract4b_new/
Enter Your Details
Name: Hina
Age: 20
Email: hina123gmail.com
Gender:
Male
Female
Hobbies:
Reading
Music
Sports
Submit
localhost:8080 says
Please enter a valid email
OK


Form Page
localhost:8080/pract4b_new/
Enter Your Details
Name: Hina
Age: 20
Email: hina123@gmail.com
Gender:
Male
Female
Hobbies:
Reading
Music
Sports
Submit
localhost:8080 says
Please select your gender
OK


Form Page
localhost:8080/pract4b_new/
Enter Your Details
Name: Hina
Age: 20
Email: hina123@gmail.com
Gender:
Male
Female
Hobbies:
Reading
Music
Sports
Submit
localhost:8080 says
Please select at least one hobby
OK


Result Page
localhost:8080/pract4b_new/Validate.jsp
Validated Successfully!!
Your Submitted Information
Name: Hina
Age: 20
Email: hina123@gmail.com
Gender: Female
Hobbies:
Reading
Music

**Date : / /**

**4c. Aim :** Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC.

**CODE :**

**<u>for database</u>**

```
>> create table users(username varchar(20) PRIMARY KEY, password1
        varchar(20),password2 varchar(20), email varchar(20), country varchar(20));

>> insert into users values ('Diksha','admin','admin','admin@admin.com','India');
```

---

**<u>login.html</u>**

```
<html><body><h1>Login Page</h1> <form action="Login.jsp" >

Enter User Name <input type="text" name="txtName" ><br>

Enter Password <input type="password" name="txtPass" ><br>

<input type="reset" ><input type="submit" value="LOGIN" >

</form></body></html>
```

---

**<u>Login.jsp</u>**

```
<%@page contentType="text/html" import="java.sql.*"%>

<html><body><h1>Registration JSP Page</h1>

<% String uname=request.getParameter("txtName");

String pass = request.getParameter("txtPass");

try{

Class.forName("com.mysql.jdbc.Driver");

Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb","root","root");

PreparedStatement stmt = con.prepareStatement("select password1 from users where
username=?");

stmt.setString(1, uname);

ResultSet rs = stmt.executeQuery();

if(rs.next()) {

if(pass.equals(rs.getString(1))) {
```

```
out.println("<h1> LOGIN SUCCESSFULLL </h1>");
} } else {
out.println("<h1>User Name not exist !!!!!</h1>"); %>
<jsp:include page="register.html" ></jsp:include>
<% } } catch(Exception e){out.println(e);}
%> </body></html>
```

---

**register.html**

```
<html><head><title>New User Registration Page</title></head>
<body><form action="Register.jsp" > <h1> New User Registration Page</h1>
Enter User Name <input type="text" name="txtName" ><br>
Enter Password <input type="password" name="txtPass1" ><br>
Re-Enter Password<input type="password" name="txtPass2" ><br>
Enter Email<input type="text" name="txtEmail" ><br>
Enter Country Name <input type="text" name="txtCon" ><br>
<input type="reset" ><input type="submit" value="REGISTER" >
</form></body></html>
```

---

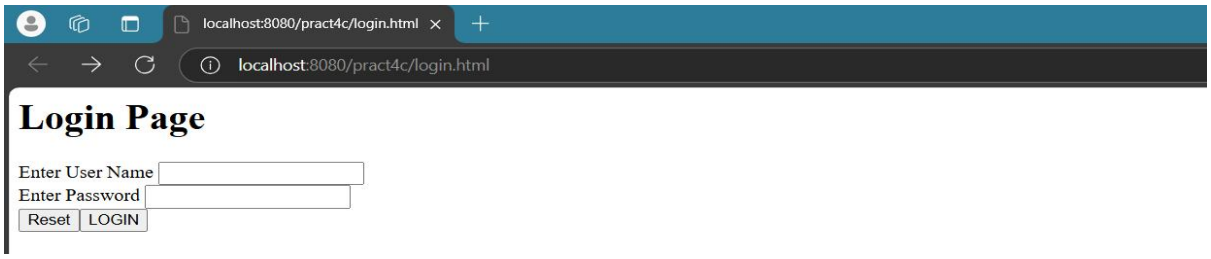
**Register.jsp**

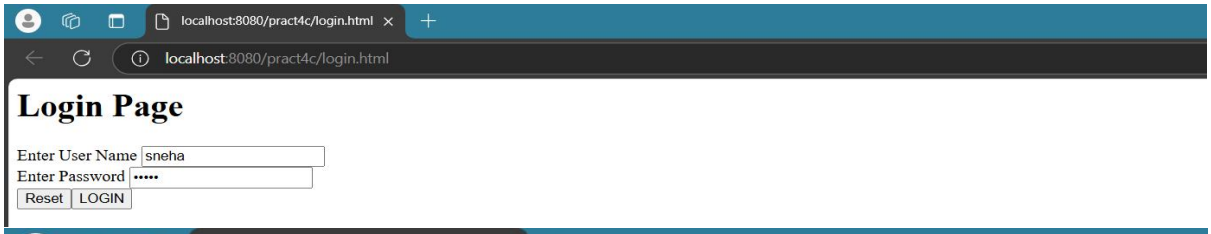
```
<%@page contentType="text/html" import="java.sql.*"%>
<html><body><h1>Registration JSP Page</h1>
<% String uname=request.getParameter("txtName");
String pass1 = request.getParameter("txtPass1");
String pass2 = request.getParameter("txtPass2");
String email = request.getParameter("txtEmail");
String ctry = request.getParameter("txtCon");
if(pass1.equals(pass2)){
try{
Class.forName("com.mysql.jdbc.Driver");
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb","root","root");
```

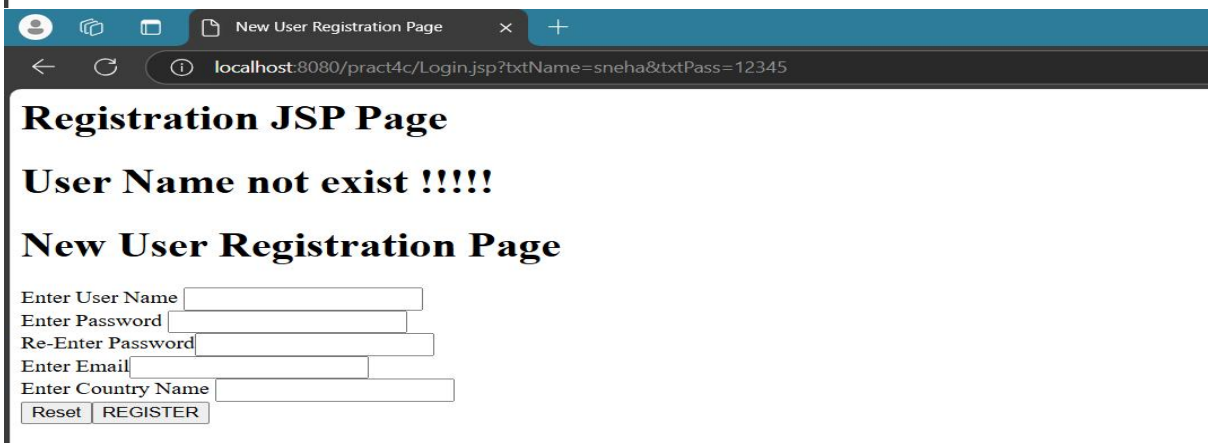
```
PreparedStatement stmt = con.prepareStatement("insert into users values (?,?,?,?,?)");
stmt.setString(1, uname); stmt.setString(2, pass1); stmt.setString(3, pass1);
stmt.setString(4, email); stmt.setString(5, ctry);
int row = stmt.executeUpdate();
if(row==1) { out.println("Registration Successful"); }
else {
out.println("Registration FAIL !!!!");
%><jsp:include page="register.html" ></jsp:include>
<% } } catch(Exception e){out.println(e);}
} else {
out.println("<h1>Password Mismatch</h1>"); %>
<jsp:include page="register.html" ></jsp:include>
<% } %> </body> </html>
```

---

**Output :**



New User Registration Page

localhost:8080/pract4c/Login.jsp?txtName=sneha&txtPass=12345

# Registration JSP Page

# User Name not exist !!!!!

# New User Registration Page

Enter User Name sneha
Enter Password •••••
Re-Enter Password •••••
Enter Email sneha@gmail.com
Enter Country Name India
Reset REGISTER

New User Registration Page

localhost:8080/pract4c/Register.jsp?txtName=sneha&txtPass1=54321&txtPass2=12345&txtEmail=sneha%40gmail.com&txtCon=India

# Registration JSP Page

# Password Mismatch

# New User Registration Page

Enter User Name
Enter Password
Re-Enter Password
Enter Email
Enter Country Name
Reset REGISTER

New User Registration Page

localhost:8080/pract4c/Login.jsp?txtName=sneha&txtPass=12345

# Registration JSP Page

# User Name not exist !!!!!

# New User Registration Page

Enter User Name sneha
Enter Password •••••
Re-Enter Password •••••
Enter Email sneha@gmail.com
Enter Country Name India
Reset REGISTER

localhost:8080/pract4c/Register.j

localhost:8080/pract4c/Register.jsp?txtName=sneha&txtPass1=32145&txtPass2=32145&txtEmail=sneha%40gmail.com&txtCon=India

# Registration JSP Page

Registration Successful

# Practical No.5

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**Date : / /**

**5a. Aim :** Create an html page with fields, eno, name, age, desg, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.

**CODE :**

**<u>for database</u>**

```
>> use logindb;
>> create table emp (empid varchar(10) PRIMARY KEY, ename varchar(50), salary
      varchar(50), age varchar(50) );
>> insert into emp values ('1', 'aaa', '221234', '11');
>> insert into emp values ('2', 'bbb', '334567', '22');
>> insert into emp values ('3', 'ccc', '44454', '33');
>> insert into emp values ('4', 'ddd', '55123', '44');
```

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**<u>index.html</u>**

```
<html><body><form action="UpdateEmp.jsp" >
Enter Employee Number<input type="text" name="txtEno" ><br>
Enter Name<input type="text" name="txtName" ><br>
Enter age<input type="text" name="txtAge" ><br>
Enter Salary<input type="text" name="txtSal" ><br>
<input type="reset" ><input type="submit">
</form></body></html>
```

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**<u>UpdateEmp.jsp</u>**

```
<%@page contentType="text/html" import="java.sql.*" %>
<html><body> <h1>Employee Record Update</h1> <%
String eno=request.getParameter("txtEno");
String name=request.getParameter("txtName");
String age = request.getParameter("txtAge");
```

```
String sal = request.getParameter("txtSal");
try{
Class.forName("com.mysql.jdbc.Driver");
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb","root","root");
PreparedStatement stmt = con.prepareStatement("select * from emp where empid=?");
stmt.setString(1, eno);
ResultSet rs = stmt.executeQuery();
if(rs.next()){
out.println("<h1>~~~ Employee "+name+" Exist ~~~ </h1>");
PreparedStatement pst1= con.prepareStatement("update emp set salary=? where empid=?");
PreparedStatement pst2= con.prepareStatement("update emp set age=? where empid=?");
pst1.setString(1, sal); pst1.setString(2, eno);
pst2.setString(1, age); pst2.setString(2, eno);
pst1.executeUpdate(); pst2.executeUpdate();
} else {
out.println("<h1>Employee Record not exist !!!!!</h1>");
}
}catch(Exception e){out.println(e);}
%></body></html>
```

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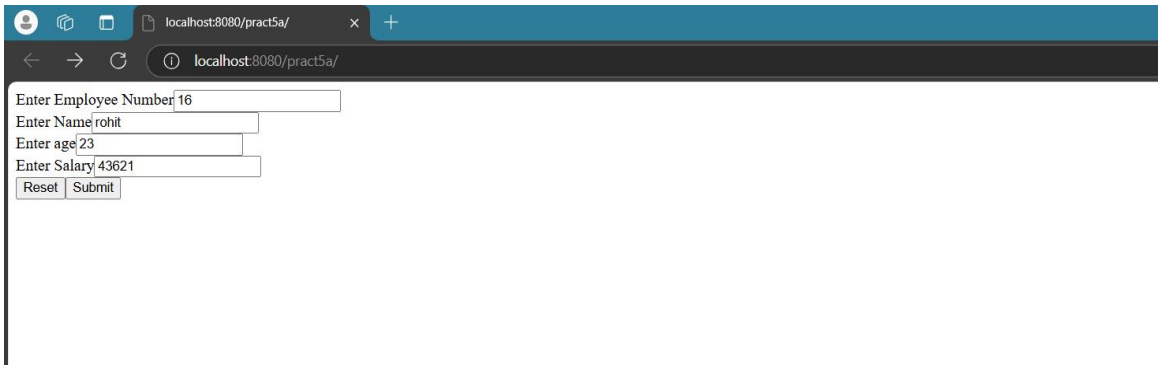
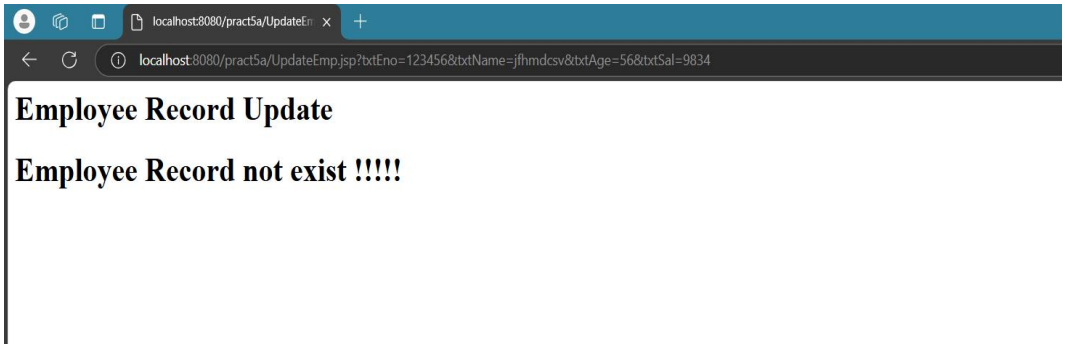
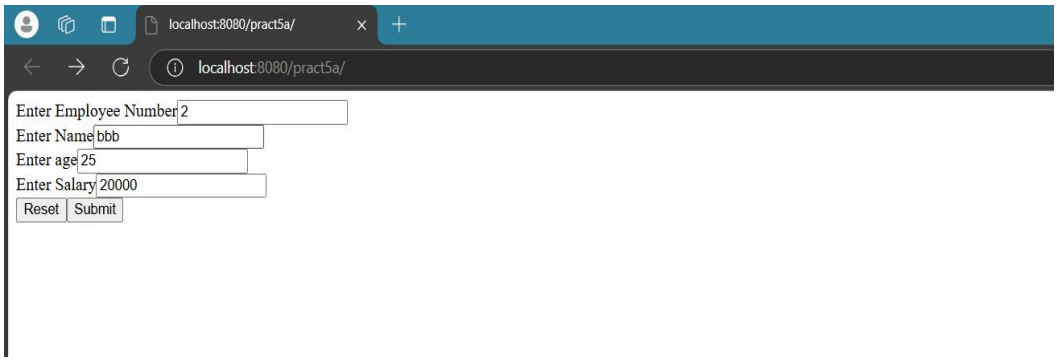
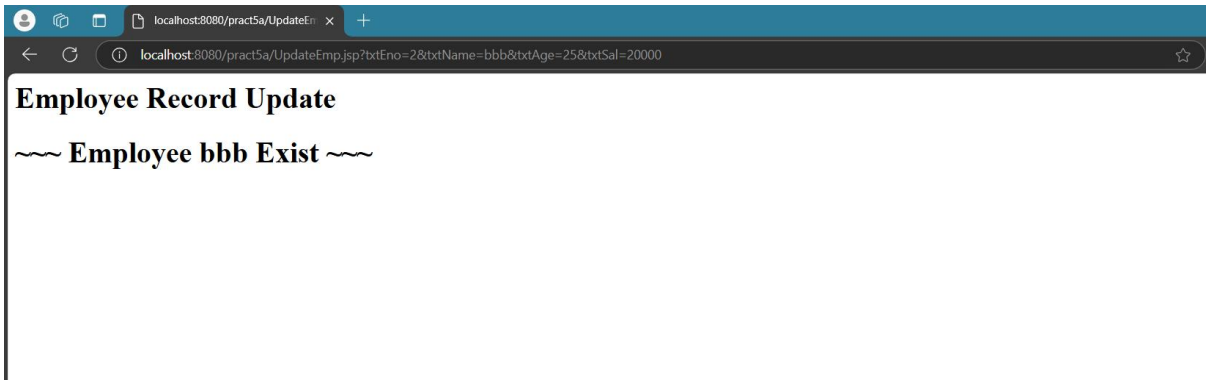
**Output :**




localhost:8080/pract5a/UpdateEm
localhost:8080/pract5a/UpdateEmp.jsp?txtEno=123456&txtName=jfhmdcsv&txtAge=56&txtSal=9834
Employee Record Update
Employee Record not exist !!!!!


localhost:8080/pract5a/
localhost:8080/pract5a/
Enter Employee Number 2
Enter Name bbb
Enter age 25
Enter Salary 20000
Reset
Submit


localhost:8080/pract5a/UpdateEm
localhost:8080/pract5a/UpdateEmp.jsp?txtEno=2&txtName=bbb&txtAge=25&txtSal=20000
Employee Record Update
~~~ Employee bbb Exist ~~~
~~~

**Date : / /**

**5b. Aim :** Create a JSP page to demonstrate the use of Expression language.

**CODE :**

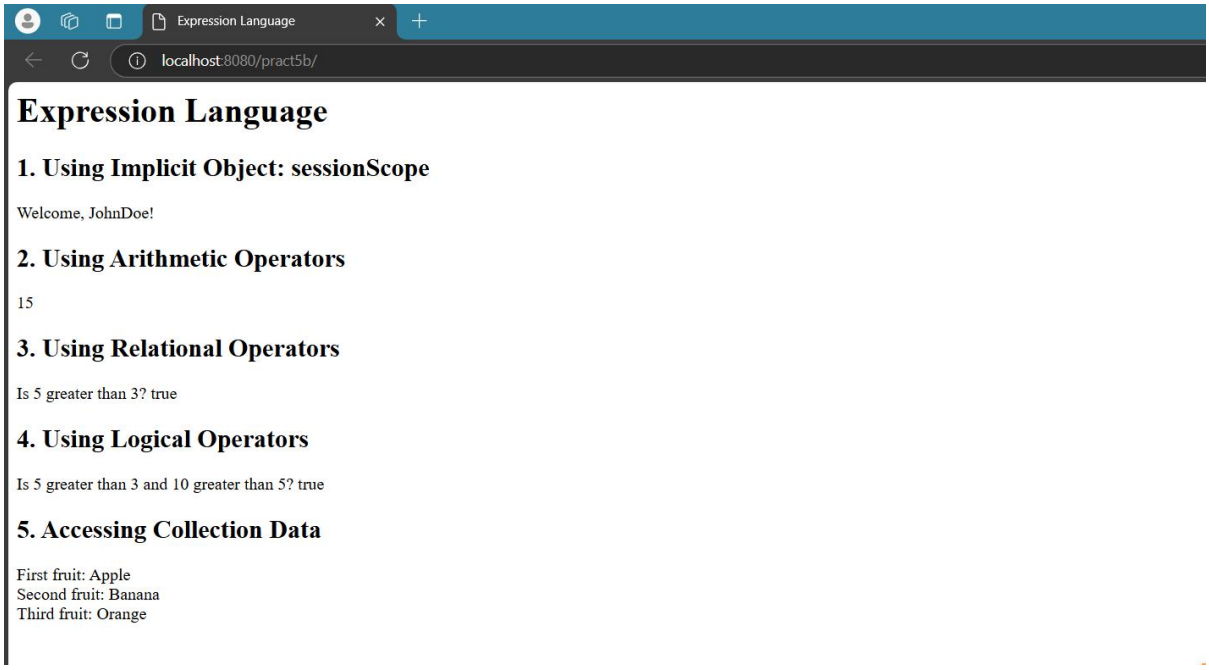
**index.jsp**

```
<%@page import="java.util.List"%>
<%@page import="java.util.Arrays"%>
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html><head><title>Expression Language</title></head>
<body><h1>Expression Language</h1>
<h2>1. Using Implicit Object: sessionScope</h2>
<%-- Setting session attribute --%>
<%
session.setAttribute("username", "JohnDoe");
%>
Welcome, ${sessionScope.username}!
<h2>2. Using Arithmetic Operators</h2>
<%-- EL Arithmetic: Adding two numbers --%>
${5 + 10}
<h2>3. Using Relational Operators</h2>
<%-- EL Relational: Comparing two numbers --%>
Is 5 greater than 3? ${5 > 3}
<h2>4. Using Logical Operators</h2>
<%-- EL Logical: Combining two conditions --%>
Is 5 greater than 3 and 10 greater than 5? ${5 > 3 && 10 > 5}
<h2>5. Accessing Collection Data</h2>
<%-- EL with a List stored in application scope --%>
<%
List<String> fruits = Arrays.asList("Apple", "Banana", "Orange");
application.setAttribute("fruitList", fruits);
```

```
%>
First fruit: ${applicationScope.fruitList[0]}<br/>
Second fruit: ${applicationScope.fruitList[1]}<br/>
Third fruit: ${applicationScope.fruitList[2]}
</body></html>
```

---

**Output :**



**Date : / /**

**5c. Aim :** Create a JSP application to demonstrate the use of JSTL.

**CODE :**

**index.html**

```
<!DOCTYPE html>
<html> <head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>JSP Page</title> </head> <body>
<h1>JSTL TAGS</h1>
<a href="jstl.jsp">Click here</a>
</body> </html>
```

---

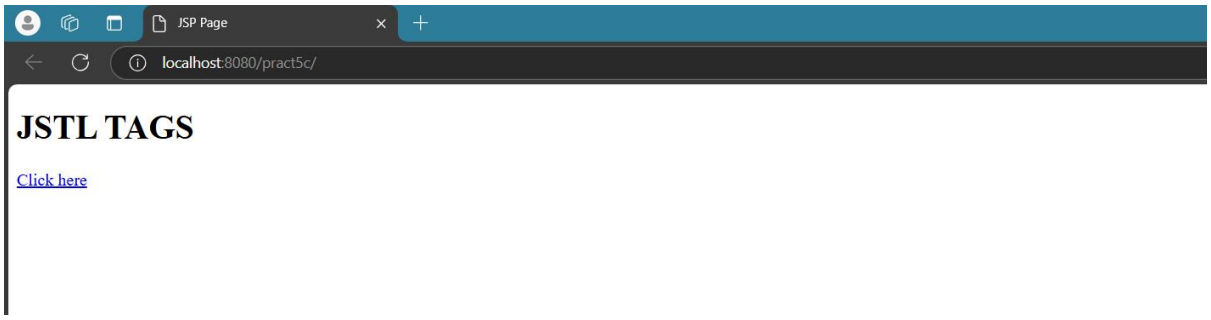
**jstl.jsp**

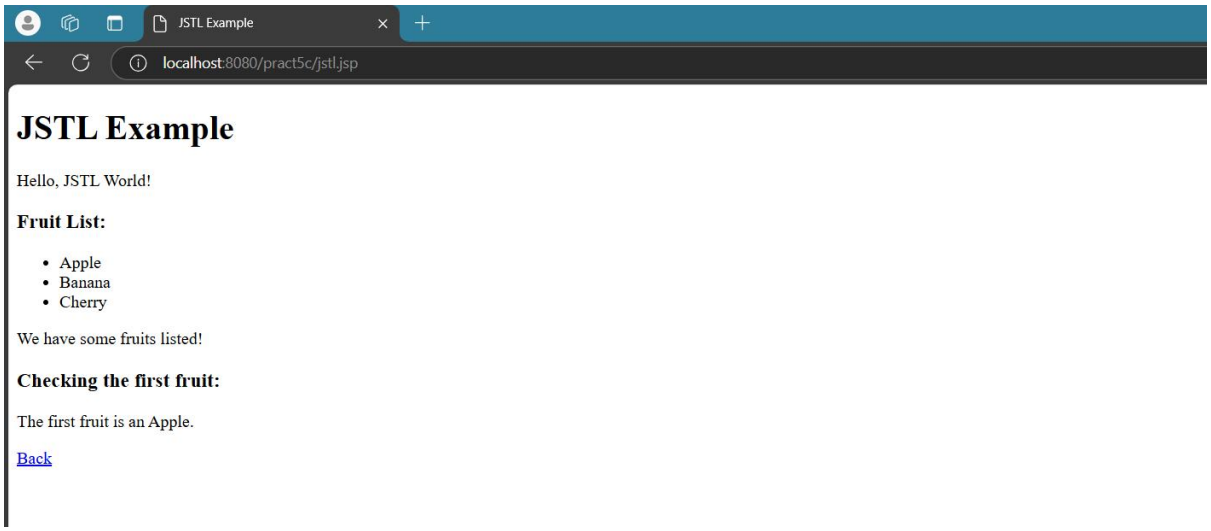
```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<!DOCTYPE html>
<html lang="en"> <head> <meta charset="UTF-8">
<title>JSTL Example</title> </head>
<body> <h1>JSTL Example</h1>
<c:set var="message" value="Hello, JSTL World!" />
<p>${message}</p>
<c:set var="fruits" value="${['Apple', 'Banana', 'Cherry']}" />
<h3>Fruit List:</h3>
<ul>
   <c:forEach var="fruit" items="${fruits}">
       <li>${fruit}</li>
  </c:forEach>
</ul>
<c:if test="${not empty fruits}">
```

```
    <p>We have some fruits listed!</p>
</c:if>
<h3>Checking the first fruit:</h3>
<c:choose>
    <c:when test="${fruits[0] == 'Apple'}">
        <p>The first fruit is an Apple.</p>
    </c:when>
    <c:otherwise>
        <p>The first fruit is not an Apple.</p>
    </c:otherwise>
</c:choose>
<a href="index.html">Back</a> <!-- Link back to index -->
</body> </html>
```

---

**Output :**



# Practical No.6

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**Date : / /**

**6a. Aim :** Create a Currency Converter application using EJB.

**CODE :**

**index.html**

```
<html><head><title>Currency Converter</title></head><body>
<form action="CCServlet" >
Enter Amount <input type="text" name="amt"><br>
Select Conversion Type
<input type="radio" name="type" value="r2d" checked>Rupees to Dollar
<input type="radio" name="type" value="d2r" >Dollar to Rupees<br>
<input type="reset" ><input type="submit" value="CONVERT" >
</form></body>
```

---

**CCServlet.java**

```
package mypack;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.ejb.EJB;
import mybeans.CCBean;
public class CCServlet extends HttpServlet {
   @EJB CCBean obj;
   public void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
            response.setContentType("text/html;charset=UTF-8");
            PrintWriter out = response.getWriter();
            double amt = Double.parseDouble(request.getParameter("amt"));
```

```
            if(request.getParameter("type").equals("r2d")) {
                out.println("<h1>"+amt+ " Rupees = "+obj.r2Dollar(amt)+" Dollars</h1>");
            }
            if(request.getParameter("type").equals("d2r")) {
                out.println("<h1>"+amt+ " Dollars = "+obj.d2Rupees(amt)+" Rupees</h1>");
            }
    }
}
```

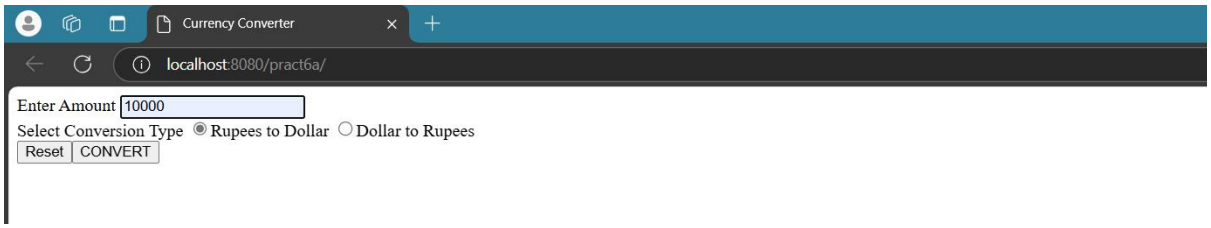
---

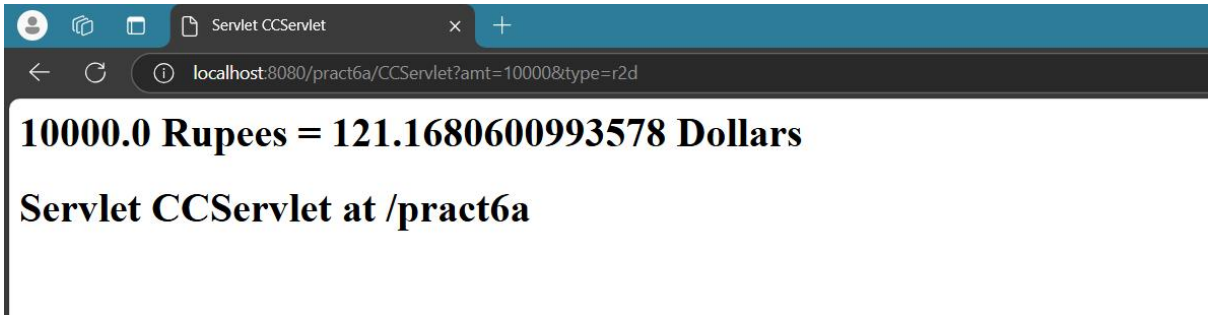
**CCBean.java**

```
package mybeans;
import javax.ejb.Stateless;
@Stateless
public class CCBean {
    public CCBean(){}
    public double r2Dollar(double r){ return r/82.53; }
    public double d2Rupees(double d){ return d*82.53; }
}
```

---

**Output :**



**Date : / /**

**6b. Aim :** Develop a Simple Room Reservation System Application Using EJB.

**CODE :**

**for database**

```
>> create database rrdb;
>> use rrdb;
>>  Create table roombook (RoomId varchar(4) PRIMARY KEY, RoomType varchar(20),
      charges double(5,2), cust varchar(20), mob varchar(20), status varchar(10));
>> insert into roombook values('1001','Delux',500.00,",",'Not Booked');
>> insert into roombook values('1002','Super Delux',700.00,",",'Not Booked');
>> insert into roombook values('1003','Suit',950.00,",",'Not Booked');
>> insert into roombook values('2001','Delux',500.00,",",'Not Booked');
>> insert into roombook values('2002','Super Delux',700.00,",",'Not Booked');
>> insert into roombook values('2003','Suit',950.00,",",'Not Booked');
```

---

**index.html**

```
<html><head><title>Room Reservation System</title></head>
<body><form action="RBServlet" >
Select a room Type
<input type="radio" name="txtType" value="Delux">Delux
<input type="radio" name="txtType" value="Super Delux">Super Delux
<input type="radio" name="txtType" value="Suit">Suit<br>
Enter Your Name<input type="text" name="txtCust" ><br>
Enter Mobile No.<input type="text" name="txtMob" ><br>
<input type="reset" ><input type="submit" value="Book Room">
</form></body></html>
```

---

**RBServlet.java**

```
package mypack;
import java.io.*;
```

```
import javax.servlet.*;
import javax.servlet.http.*;
import javax.ejb.EJB;
import mybeans.RRBean;
public class RBServlet extends HttpServlet {
   @EJB RRBean obj;
   public void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException{
            PrintWriter out=response.getWriter();
            String rt=request.getParameter("txtType");
            String cn=request.getParameter("txtCust");
            String cm=request.getParameter("txtMob");
            String msg = obj.roomBook(rt, cn, cm);
            out.println(msg);
   }
}
```

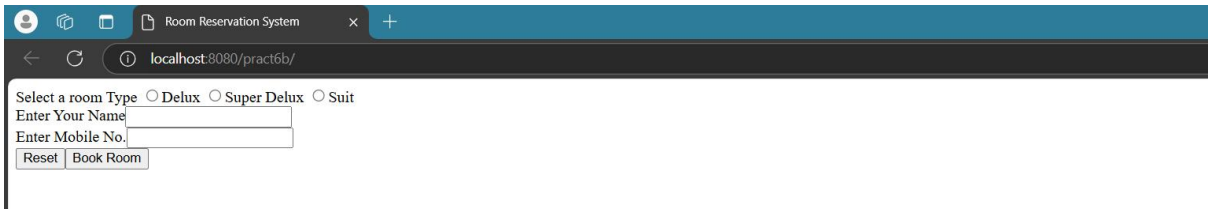
---

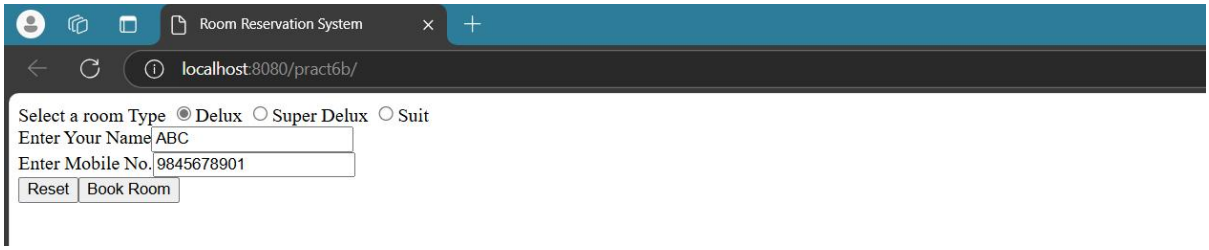
**RRBean.java**

```
package mybeans;
import javax.ejb.Stateless;
import java.sql.*;
@Stateless
public class RRBean {
   public RRBean(){}
   public String roomBook(String rt, String cn, String cm){
   String msg="";
   try{
            Class.forName("com.mysql.jdbc.Driver");
            Connection con =
   DriverManager.getConnection("jdbc:mysql://localhost:3306/rrdb","root","root");
```

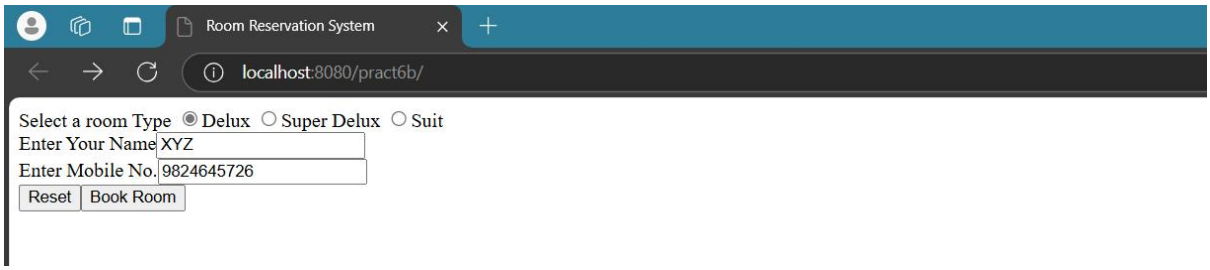
```
            String query="select * from roombook where RoomType=? and status='Not
Booked'";
            PreparedStatement pst = con.prepareStatement(query);
            pst.setString(1,rt);
            ResultSet rs= pst.executeQuery();
            if(rs.next()){
                    String rno=rs.getString(1);
                    PreparedStatement stm1 = con.prepareStatement("update roombook set
cust=? where RoomId=? ");
                    PreparedStatement stm2 = con.prepareStatement("update roombook set
mob=? where RoomId=? ");
                    PreparedStatement stm3 = con.prepareStatement("update roombook set
status=? where RoomId=? ");
                    stm1.setString(1,cn); stm1.setString(2,rno);
                    stm2.setString(1,cm); stm2.setString(2,rno);
                    stm3.setString(1, "Booked"); stm3.setString(2,rno);
                    stm1.executeUpdate();
                    stm2.executeUpdate();
                    stm3.executeUpdate();
                    msg = "Room "+rno+ " Booked <br> Charges = "+rs.getString(3);
            } else {
                    msg = "Room "+rt+ " currently Not available";
            }
   }catch(Exception e){msg=""+e;}
            return msg;
  }
}
```

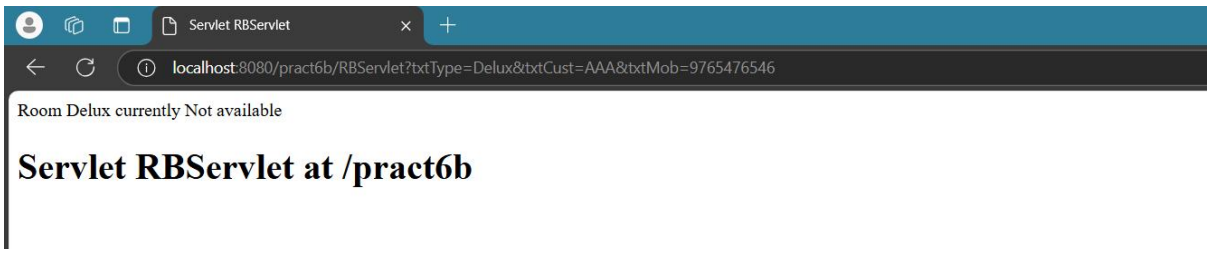
**Output :**


Room Reservation System
localhost:8080/pract6b/
Select a room Type Delux Super Delux Suit
Enter Your Name
Enter Mobile No.
Reset Book Room


Room Reservation System
localhost:8080/pract6b/
Select a room Type Delux Super Delux Suit
Enter Your Name ABC
Enter Mobile No. 9845678901
Reset Book Room

Servlet RBServlet
localhost:8080/pract6b/RBServlet?txtType=Delux&txtCust=ABC&txtMob=9845678901
Room 1001 Booked
Charges = 500.00
Servlet RBServlet at /pract6b


Room Reservation System
localhost:8080/pract6b/
Select a room Type Delux Super Delux Suit
Enter Your Name XYZ
Enter Mobile No. 9824645726
Reset Book Room


Servlet RBServlet
localhost:8080/pract6b/RBServlet?txtType=Delux&txtCust=AAA&txtMob=9765476546
Room Delux currently Not available
Servlet RBServlet at /pract6b

**Date : / /**

**6c. Aim :** Develop simple shopping cart application using EJB [Stateful Session Bean].

**CODE :**

**index.html**

```
<!DOCTYPE html>
<html lang="en">
<head> <meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Shopping Cart</title> </head>
<body> <h1>Shopping Cart</h1>
<form action="cart" method="post">
<input type="text" name="item" placeholder="Enter item" required>
<button type="submit" name="action" value="add">Add Item</button>
<button type="submit" name="action" value="remove">Remove Item</button>
<button type="submit" name="action" value="clear">Clear Cart</button> </form>
<form action="cart" method="get">
<button type="submit">View Cart</button> </form>
</body> </html>
```

---

**ShoppingCartServlet.java**

```
package SCart;
import java.io.IOException;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/cart")
```

```
public class ShoppingCartServlet extends HttpServlet {
    @EJB
    private ShoppingCartBean cart;
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String action = request.getParameter("action");
        String item = request.getParameter("item");
        if ("add".equals(action)) {
            cart.addItem(item);
        } else if ("remove".equals(action)) {
            cart.removeItem(item);
        } else if ("clear".equals(action)) {
            cart.clearCart();
        }
        response.getWriter().println("Cart Contents: " + cart.getItems());
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.getWriter().println("Cart Contents: " + cart.getItems());
    }
}
```

---

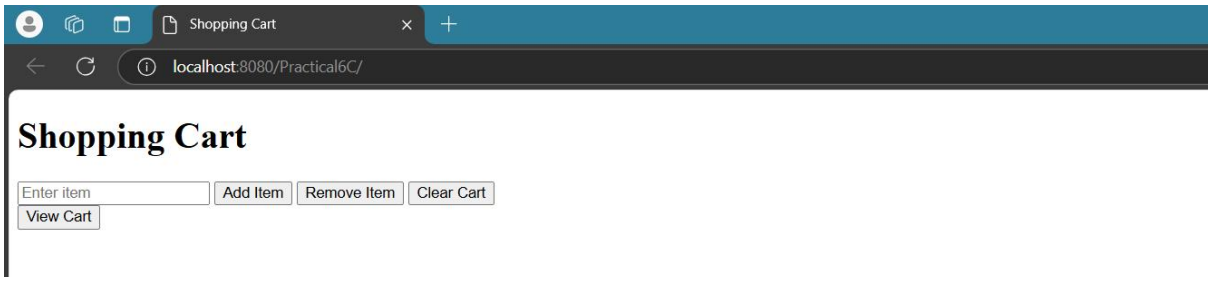
**ShoppingCartBean.java**

```
package SCart;
import java.util.ArrayList;
import java.util.List;
import javax.ejb.Stateful;
```

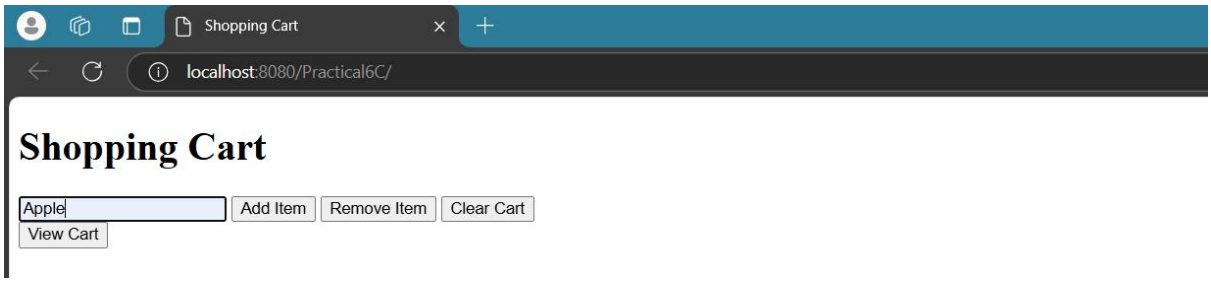
```
@Stateful
public class ShoppingCartBean {
    private List<String> items;
    public ShoppingCartBean() {
        items = new ArrayList<>();
    }
    public void addItem(String item) {
        items.add(item);
    }
    public void removeItem(String item) {
        items.remove(item);
    }
    public List<String> getItems() {
        return items;
    }
    public int getItemCount() {
        return items.size();
    }
    public void clearCart() {
        items.clear();
    }
}
```

---

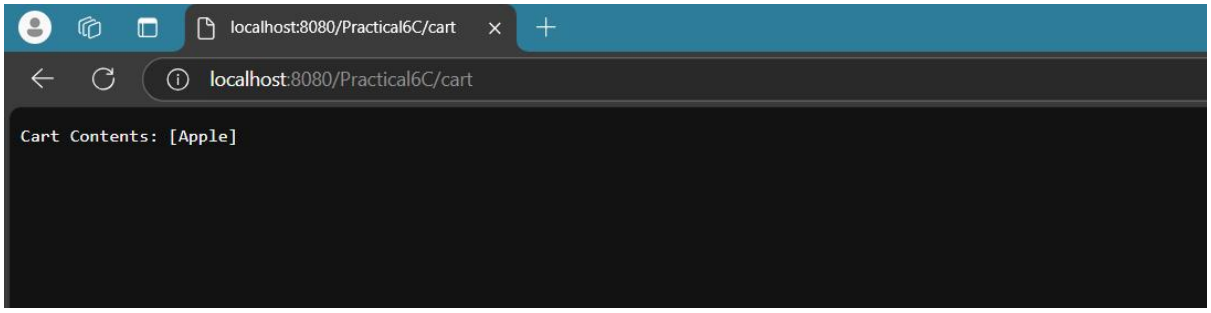
**Output :**



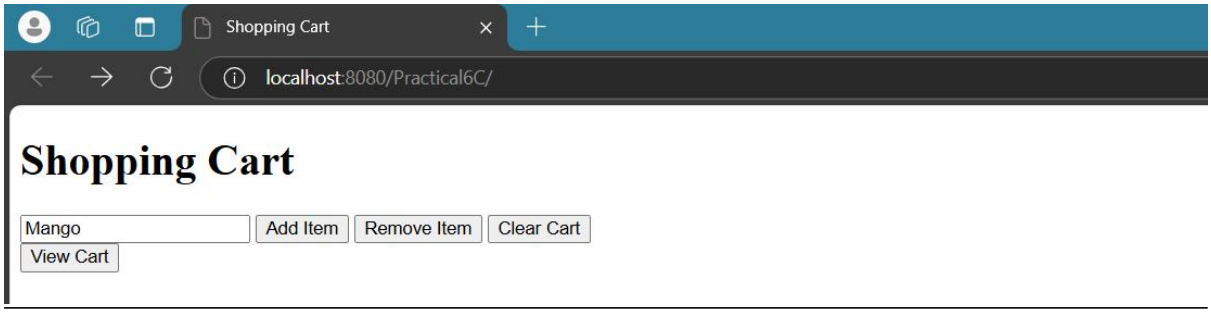
Adding Apple



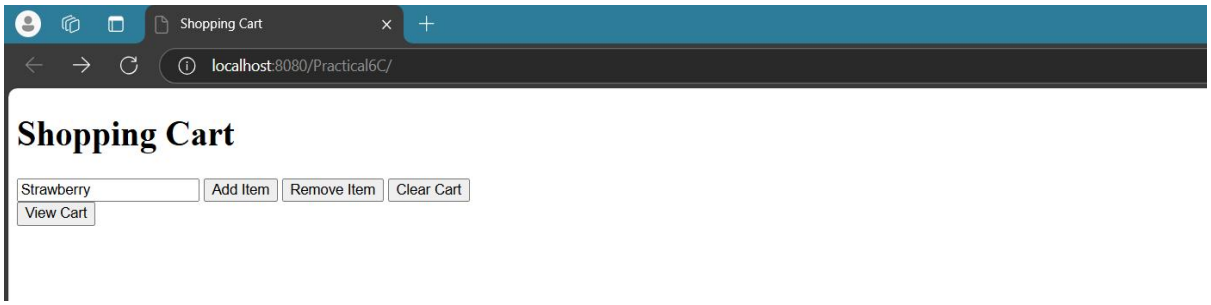

Viewing



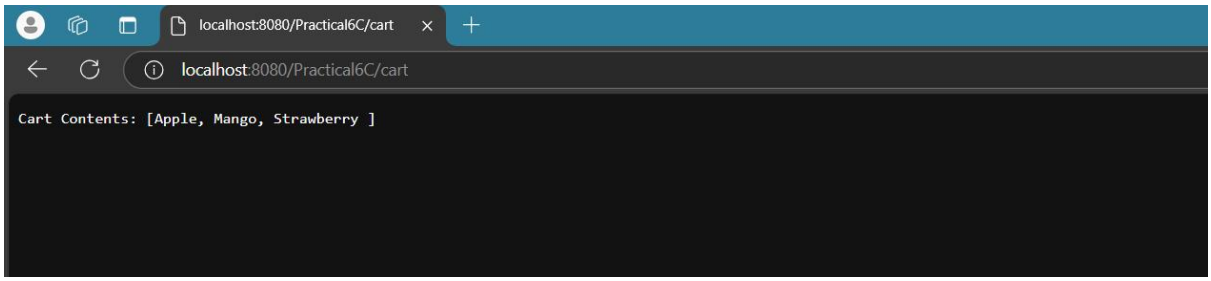

Adding Mango

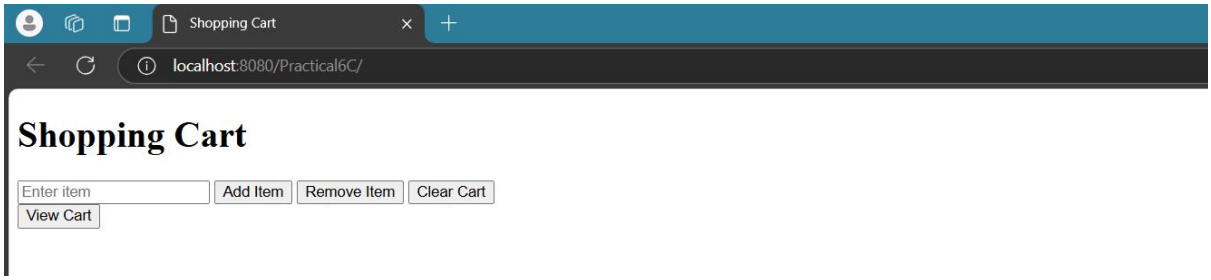



Adding Strawberry

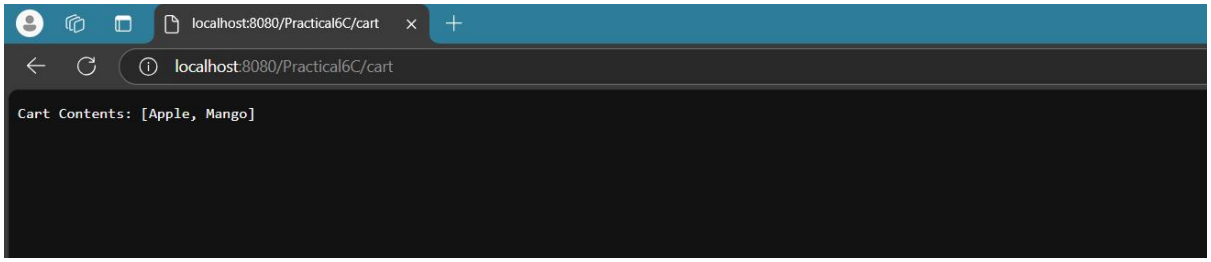



Viewing the Cart

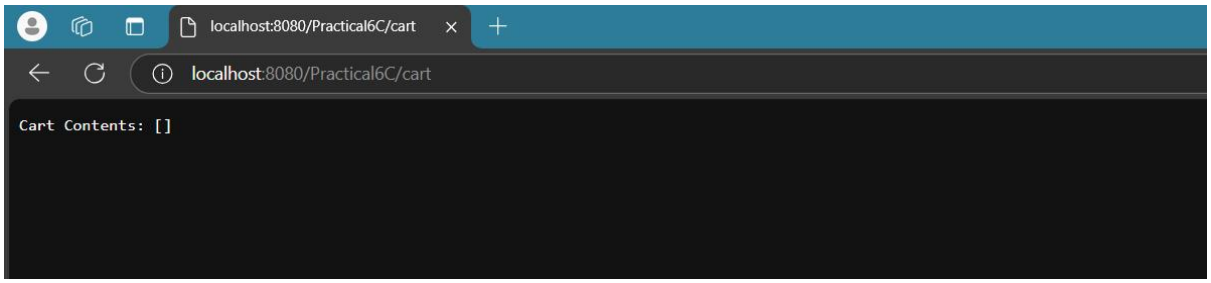





Removing an Item




Clearing the Cart



# Practical No.7

---

**Date : / /**

**7a. Aim :** Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

**CODE :**

**index.html**

```
<html> <head> <meta http-equiv="Refresh" content="0; URL=ServletClient">
</head> <body> </body> </html>
```

---

**ServletClient.java**

```
package servlet;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name= "ServletClient", urlPatterns={"/ServletClient"})
public class ServletClient extends HttpServlet {
  @EJB CounterServletHitsBean CounterBean;
  @Override
  protected void service(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        out.println("<!DOCTYPE html>");
```

```
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet ServletClient</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Number of times this servlet is accessed:" +
          CounterBean.incrementAndGetHitCount()+"</h1>");
        out.println("</body>");
        out.println("</html>");
    } } }
```

---

**CounterServletHitsBean.java**

```
package servlet;
import javax.ejb.Singleton;
@Singleton
public class CounterServletHitsBean {
    private int hitcount;
    public int incrementAndGetHitCount() {
            return hitcount++;
    } }
```

---

**Output :**

**Date : / /**

**7b. Aim :** Develop simple visitor Statistics application using Message Driven Bean [Stateless Session Bean].

**CODE :**

**index.jsp**

```
<%@page import="javax.jms.JMSException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Session"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!  private static ConnectionFactory connectionFactory;
private static Queue queue;
Connection connection=null;
Session mySession=null;
MessageProducer messageProducer=null;
TextMessage message=null; %>
<html><head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>JSP Page</title></head>
<body> Welcome to My Home Page <%
try{
InitialContext ic= new InitialContext();
queue= (Queue)ic.lookup("jms/Queue");
connectionFactory=(ConnectionFactory)ic.lookup("jms/QueueFactory");
connection= connectionFactory.createConnection();
```

```
mySession=connection.createSession(false, Session.AUTO_ACKNOWLEDGE);
messageProducer=mySession.createProducer(queue);
message=mySession.createTextMessage();
message.setText(request.getRemoteAddr());
messageProducer.send(message); }
catch(JMSException e) { System.out.println("Exception Occoured "+e.toString()); }  %>
</body></html>
```

---

**VisitorStatBean.java**

```
package ejb;
import java.sql.*;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VisitorStatBean {
private Connection conn=null;
private ResultSet rs;
private Statement st=null;
private String query =null;
@PostConstruct
public void connect() {
try {
Class.forName("com.mysql.jdbc.Driver").newInstance();
conn=DriverManager.getConnection("jdbc:mysql://localhost/visitorstat", "root", "tiger");
}
catch (Exception e) { System.err.println(e.getMessage()); }
@PreDestroy
public void disconnect() {
try {
```

```
conn.close();
} catch (Exception e) {
System.err.println(e.getMessage());
} }
public void addVisitor(String host) {
try {
st= conn.createStatement();
query="insert into userstat (hostname,visits) values ('"+host+"','1')";
st.executeUpdate(query);
}
catch (SQLException e) {
try {
st=conn.createStatement();
query="update userstat set visits=visits+1 where hostname='"+host+"' ";
st.executeUpdate(query);
}
catch (SQLException ex) {
System.err.println("Cannot Update"+e.getMessage());
} } } }
```

---

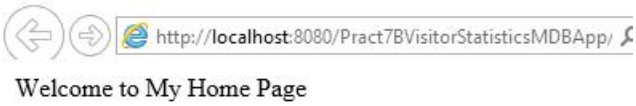
**BasicMessageBean.java**

```
package ejb;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSException;
import javax.jms.Message;
import javax.jms.MessageListener;
```

```
import javax.jms.TextMessage;
@MessageDriven(activationConfig = {
@ActivationConfigProperty(propertyName = "destinationLookup", propertyValue =
"jms/Queue"),
@ActivationConfigProperty(propertyName = "destinationType", propertyValue =
"javax.jms.Queue")
})
public class BasicMessageBean implements MessageListener {
@EJB VisitorStatBean vs;
@Resource
private MessageDrivenContext mdc;
public BasicMessageBean() {
}
@Override
public void onMessage(Message message) {
try {
if(message instanceof TextMessage){
TextMessage msg= (TextMessage) message;
vs.addVisitor(msg.getText());
} }
catch (JMSException e) {
mdc.setRollbackOnly();
} } }
```

---

**Output :**




Welcome to My Home Page

**Date : / /**

**7c. Aim :** Develop simple Marks Entry Application to demonstrate accessing Database using EJB.

**CODE :**

**for database**

```
>> create table studentdata (sname varchar(10), sub1 varchar(10), sub2 varchar(10));
>> insert into studentdata values('aaa', '50', '40');
```

---

**index.html**

```
<html> <head>
<title>TODO supply a title</title> <meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head> <body> <form action="MEServlet" >
Enter Student Name<input type="text" name="sname" ><br>
Enter Subject 1<input type="text" name="sub1"><br>
Enter Subject 2<input type="text" name="sub2" ><br>
<input type="reset" value="RE-ENTER"><input type="submit" value="SUBMIT">
</form> </body> </html>
```

---

**MEServlet.java**

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.ejb.EJB;
import mybeans.MEBean;
public class MEServlet extends HttpServlet {
    @EJB MEBean obj;
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
                String s1=request.getParameter("sub1");
                String n=request.getParameter("sname");
                String s2=request.getParameter("sub2");
                String msg = obj.markenter(n, s1, s2);
                response.setContentType("text/html;charset=UTF-8");
                try (PrintWriter out = response.getWriter()) {
                    out.println("<!DOCTYPE html>");
                    out.println("<html>");
                    out.println("<head>");
                    out.println("<title>Servlet MEServlet</title>");
                    out.println("</head>");
                    out.println("<body>");
                    out.println("<h2>"+msg+"</h2>");
                    out.println("<h1>Servlet MEServlet at " + request.getContextPath() +
"</h1>");
                    out.println("</body>");
                    out.println("</html>");
                }
    }
}
```

---

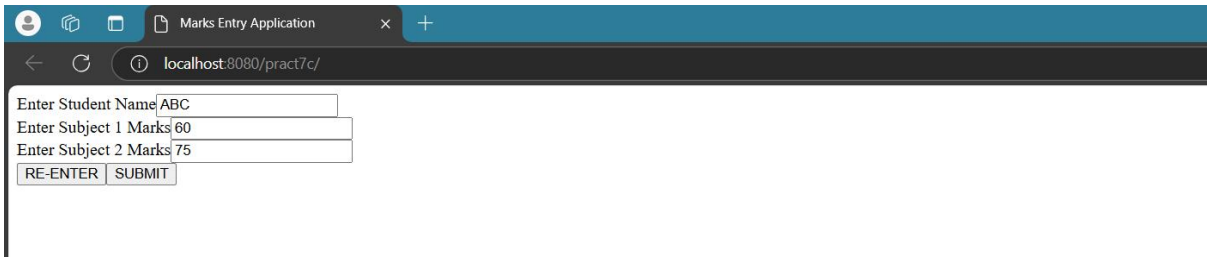
**MEBean.java**

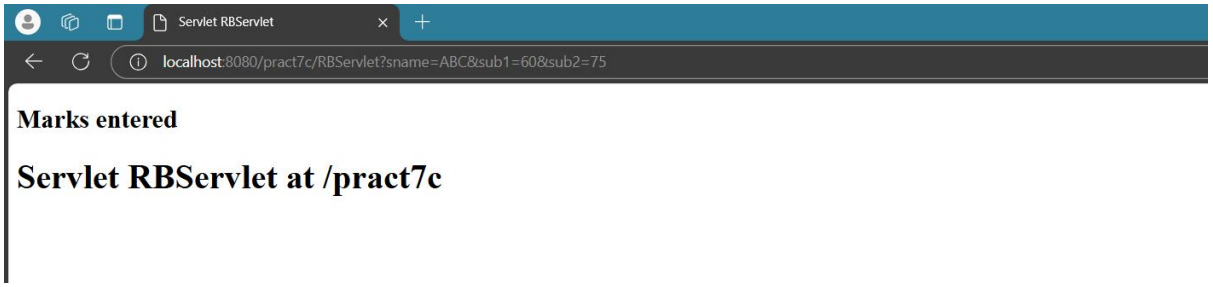
```
package mybeans;
import javax.ejb.Stateless;
import java.sql.*;
@Stateless
public class MEBean {
public MEBean(){}
  public String markenter(String n, String s1,String s2) {
```

```
    String msg;
     try{
            Class.forName("com.mysql.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/marksdb","root","root");
            String query="insert into studentdata values(?,?,?)";
            PreparedStatement pst = con.prepareStatement(query);
            pst.setString(1,n);
            pst.setString(2,s1);
            pst.setString(3,s2);
            pst.executeUpdate();
            msg="Marks entered";
     } catch(Exception e){msg=""+e;}
     return msg;
    }
}
```

---

**Output :**



# Practical No.8

---

**Date : / /**

**8a. Aim :** Develop a simple Inventory Application Using JPA.

**CODE :**

**index.html**

```
<html><body><form action="AddInventory" method="post">
<table><tr><td colspan="2" align="center"><b> Add the item in Inventory </b></td></tr>
<tr><td> Item Name : </td>
    <td> <input type="text" name="itemname" maxlength="25" size="50" /></td></tr>
<tr><td> Price : </td>
    <td> <input type="number" name="itemprice" size="50" /> </td></tr>
<tr><td> Quantity :</td>
    <td> <input type="number" name="itemquantity" /> </td></tr>
<tr><td colspan="2"> <input type="submit" name="AddToInventory"
        value="AddToInventory" /> </td></tr>
</table></form></body></html>
```

---

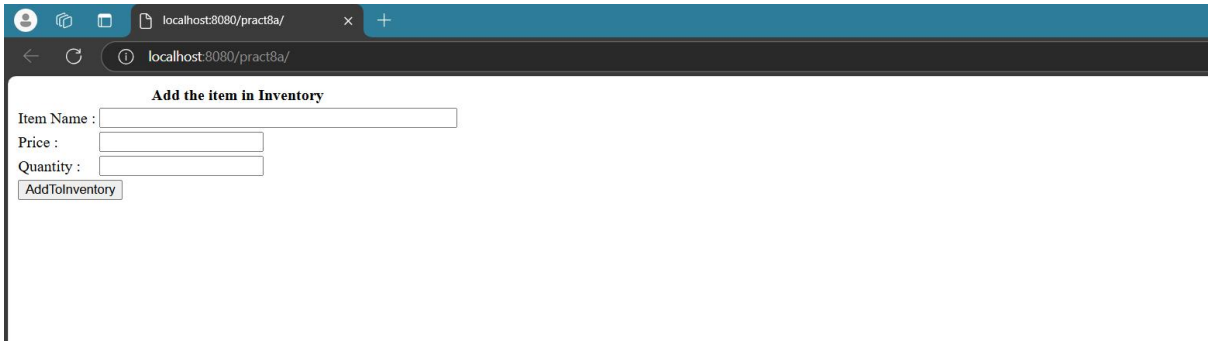
**AddInventory.java**

```
package test;
import javax.persistence.EntityManager;
import javax.persistence.EntityManagerFactory;
import javax.persistence.EntityTransaction;
import javax.persistence.Persistence;
public class AddInventory extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
     try (PrintWriter out = response.getWriter()) {
```

```
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet AddInventory</title>");
        out.println("</head>");
        out.println("<body>");
        EntityManagerFactory entityManagerFactory =
Persistence.createEntityManagerFactory("p8aPU");
        EntityManager entityManager = entityManagerFactory.createEntityManager();
        EntityTransaction entityTransaction = entityManager.getTransaction();
        Inventory il=new Inventory();
        il.setItemName(request.getParameter("itemname"));
        il.setItemQuantity(request.getParameter("itemquantity"));
        il.setItemPrice(Integer.parseInt(request.getParameter("itemprice")));
        il.setSku(Integer.parseInt(request.getParameter("SKU")));
        entityTransaction.begin();
        entityManager.persist(il);
        entityTransaction.commit();
        out.println("<h1>Record is added successfully !!! </h1>");
        out.println("</body>");
        out.println("</html>");
    } } }
```

---

**Output :**



# Practical No.9

---

**Date : / /**

**9a. Aim :** Develop a JPA Application to demonstrate use of ORM associations.

**CODE :**

**index.html**

```
<html> <head>
<title>TODO supply a title</title> <meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0"> </head>
<body>
User Details <hr><br><br>  <form action="userview.jsp" >
Name <input type="text" name="uname" maxlength="20"><br>
User Type <input type="text" name="utype" maxlength="20">
<br><input type="submit" value="submit">
</form> </body> </html>
```

---

**userview.jsp**

```
<%@page import="java.util.List"%>
<%@page import="java.util.Iterator"%>
<%@page import="hibernate.User"%>
<%!
SessionFactory sf;
org.hibernate.Session ss;
List<hibernate.User> User;
%> <%
sf = new Configuration().configure().buildSessionFactory();
ss= sf.openSession();
Transaction tx=null;
User ur=new User();
```

```
try  {
    tx=ss.beginTransaction();
    String uname=request.getParameter("uname");
    String utype=request.getParameter("utype");
    ur.setUname(uname);
    ur.setUtype(utype);
    ss.save(ur);
    tx.commit();
}  catch(Exception e){ out.println("Error"+e.getMessage()); }
try {
    ss.beginTransaction();
    User=ss.createQuery("from User").list();
} catch(Exception e){ } %>
<html> <head>
<title>Guest View</title>
</head> <body>
Guest View Click here to go <a href="index.html"> BACK </a> <br><br>
<%
Iterator it=User.iterator();
while(it.hasNext()) {
    User eachrecord=(User)it.next();
    out.print(eachrecord.getUid()+" ");
    out.print(eachrecord.getUname()+"<br>");
    out.print(eachrecord.getUtype()+"<br><hr>");
}
%>
</body> </html>
```

**hibernate.revenge.xml**

```
<hibernate-reverse-engineering>
<schema-selection match-catalog="userdb"/>
```

```
<table-filter match-name="user"/>
</hibernate-reverse-engineering>
```

---

**hibernate.cfg.xml**

```
<hibernate-configuration> <session-factory>
<property name="hibernate.dialect">org.hibernate.dialect.MySQLDialect</property>
<property name="hibernate.dialect">org.hibernate.dialect.MySQLDialect</property>
<property
name="hibernate.connection.url">jdbc:mysql://localhost:3306/userdb?zeroDateTimeBehavior=conve
rtToNull</property>
<property name="hibernate.connection.username">root</property>
<property name="hibernate.connection.password">tiger</property>
</session-factory> </hibernate-configuration>
```

---

**User.hbm.xml**

```
<hibernate-mapping>
<class optimistic-lock="version" catalog="userdb" table="user" name="hibernate.User">
<id name="uid" type="java.lang.Integer">
<column name="uid"/>
<generator class="identity"/> </id>
<property name="uname" type="string">
<column name="uname" length="20"/>
</property>
<property name="utype" type="string">
<column name="utype" length="100"/>
</property> </class>
</hibernate-mapping>
```

---

**User.java**

```
package hibernate;
```

```
public class User implements java.io.Serializable {
private Integer uid;
private String uname;
private String utype;
public User() { }
public User(String uname, String utype) {
this.uname = uname;
this.utype = utype;
}
public Integer getUid() {
return this.uid;
}
public void setUid(Integer uid) {
this.uid = uid;
}
public String getUname()  {
return this.uname;
}
public void setUname(String uname) {
this.uname = uname;
}
public String getUtype() {
return this.utype;
}
public void setUtype(String utype) {
this.utype = utype;
}  }
```

**Output :**

http://localhost:8080/WebApplication1/userview.jsp?t JSP Page JSP Page Guest View

Guest View Click here to go BACK

5
xyz
admin

---

6
pqr
user

---

7
mnq
admin

---

http://localhost:8080/WebApplication1/index.jsp JSP Page JSP Page JSP Page

Name
Type

submit

http://localhost:8080/WebApplication1/index.jsp JSP Page JSP Page JSP Page

Name xyz
Type admin

submit